

CS 31204: Computer Networks – History and Protocol Stack

Department of Computer Science
and Engineering



INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR

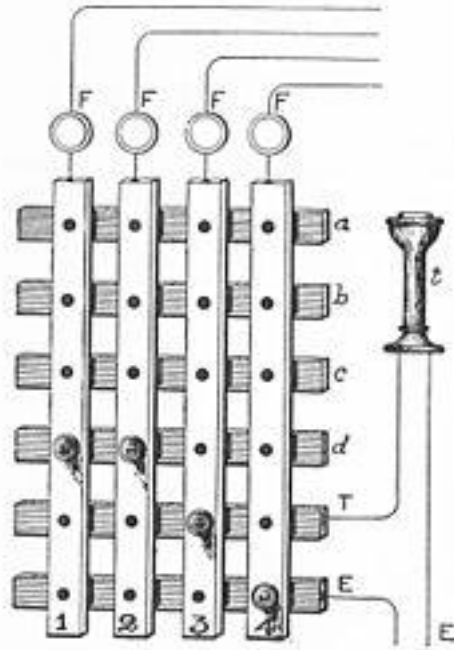
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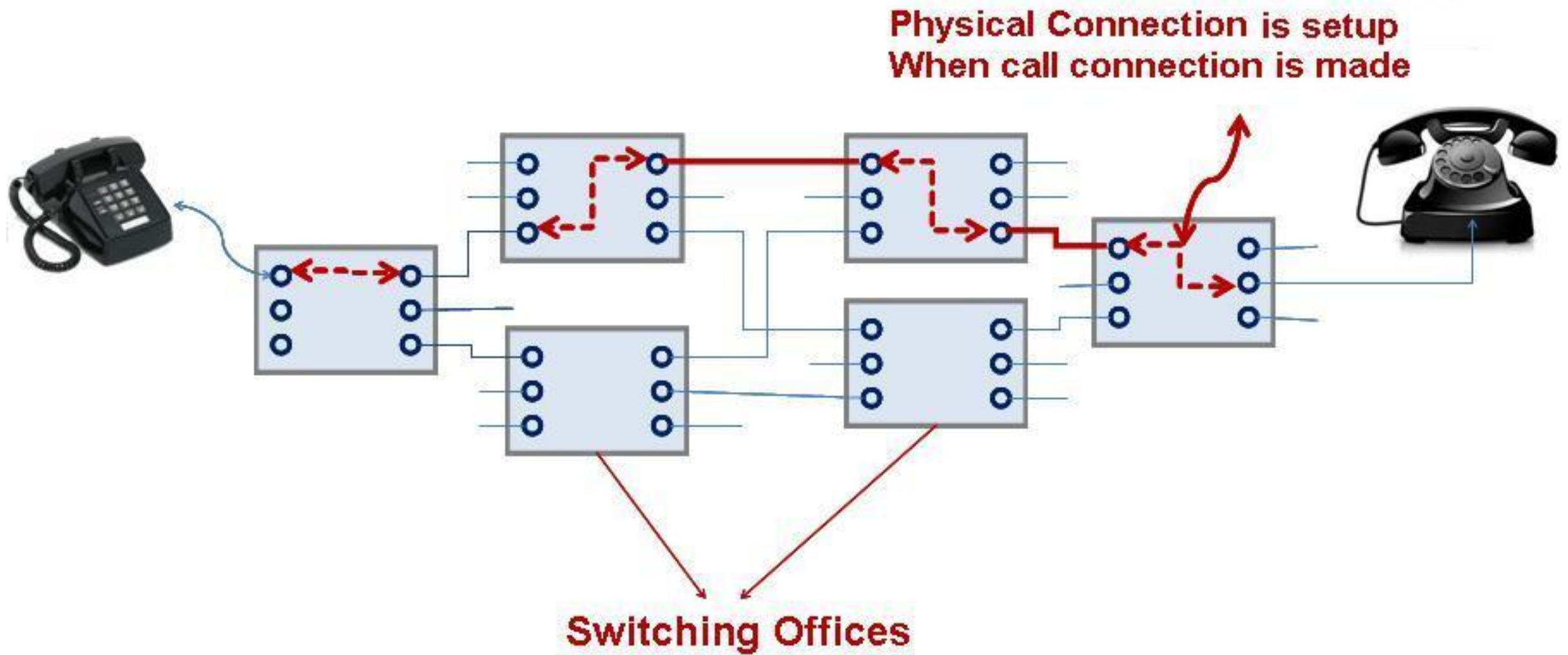


Circuit Switching – the Beginning

- The concept of how telephone switching works – creates a dedicated communication links between two communication nodes (telephones)
- In January 1878, the first telephone switch went into operation in New Haven Connecticut.



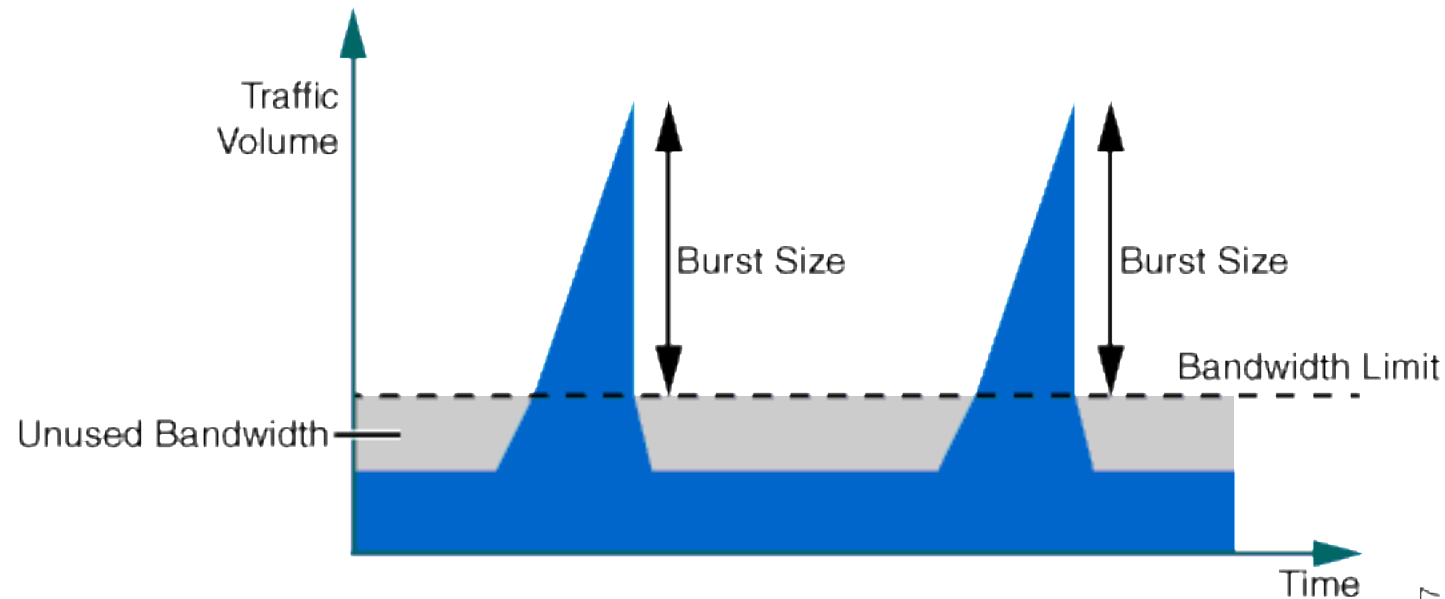
Circuit Switching



Source: <http://comp380group4project1.web.unc.edu/technical-discussion/circuit-switching/>

Problem with Circuit Switching for Data Communication

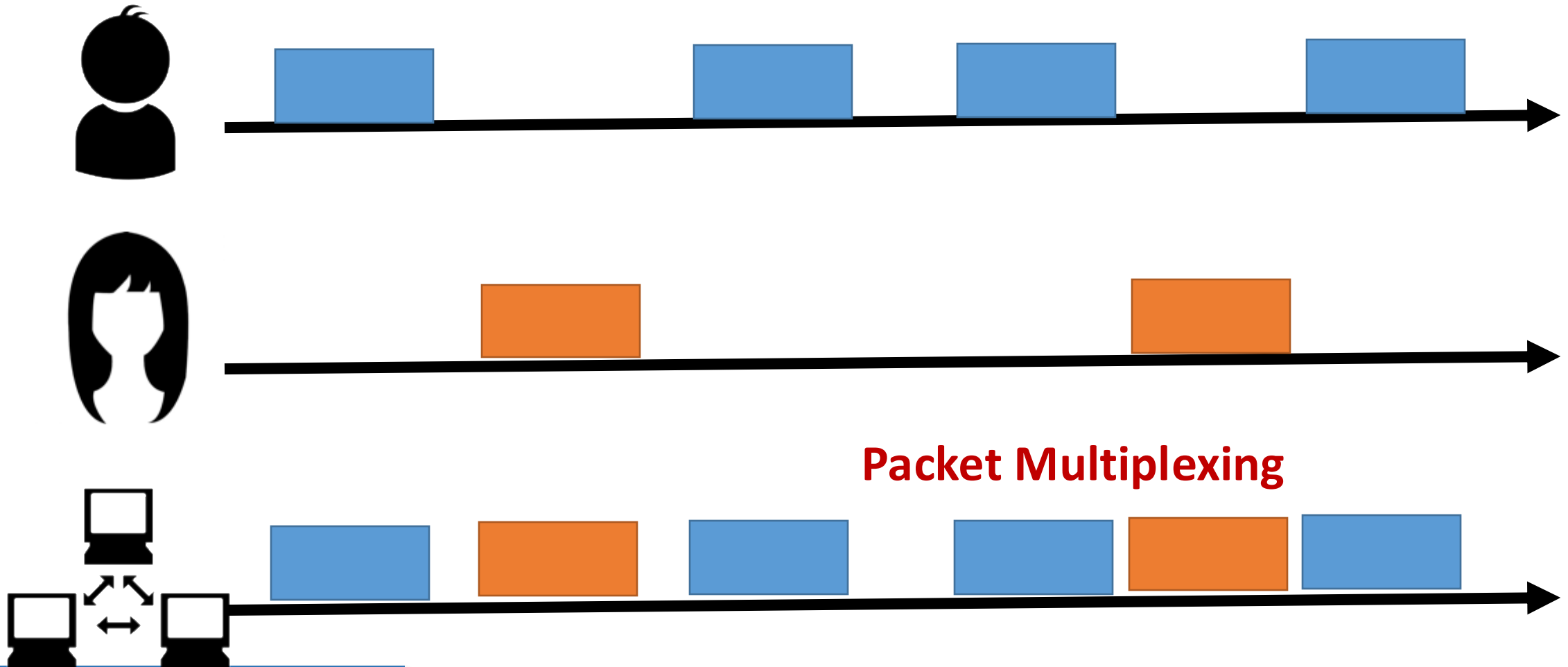
- Data traffic is bursty – uses an on-off pattern for data communication.



g041267

Packet Switching

- Decide data boundary from the communication of one user – data packets



The First Packet Switching Network

Robert Taylor was promoted to the head of the information processing office at **Defense Advanced Research Projects Agency (DARPA)** in June 1966. He intended to realize **Licklider's** ideas of an interconnected networking system. Bringing in **Larry Roberts** from MIT, he initiated a project to build such a network. The first ARPANET link was established between the **University of California, Los Angeles (UCLA)** and the **Stanford Research Institute** at 22:30 hours on October 29, 1969.

Kleinrock said in an interview: "We typed the L and we asked on the phone,

"Do you see the L?"

"Yes, we see the L," came the response.

We typed the O, and we asked, "Do you see the O."

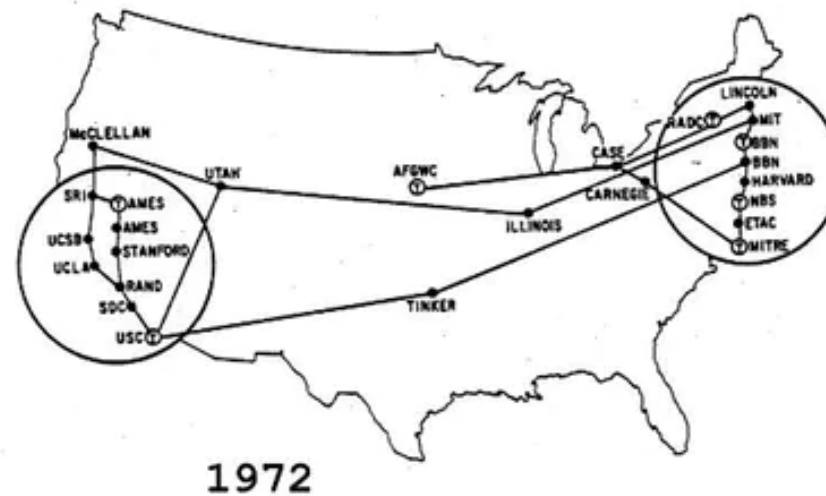
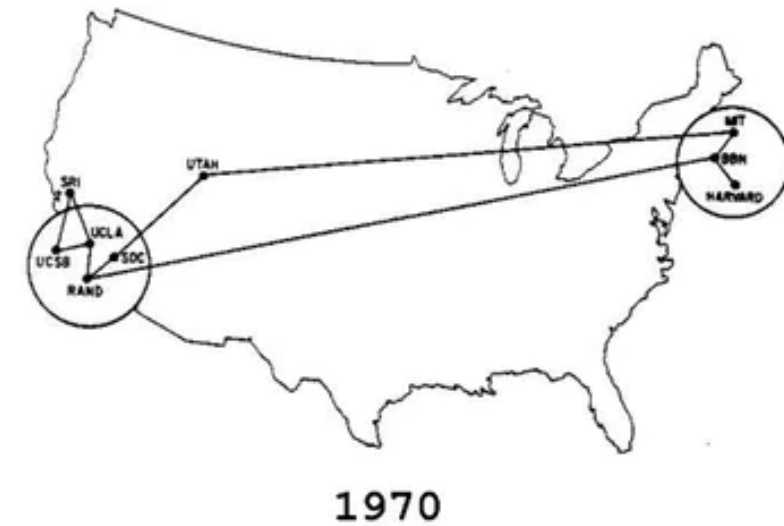
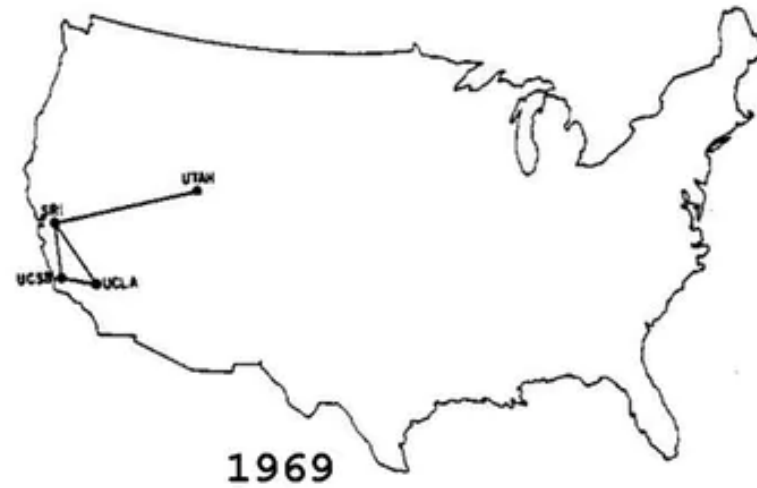
"Yes, we see the O."

Then we typed the G, and the system crashed ...

Yet a revolution had begun"

By December 5, 1969, a 4-node network was connected by adding the University of Utah and the University of California, Santa Barbara.

ARPANET – The First Packet Switching Network



History of Computer Networks

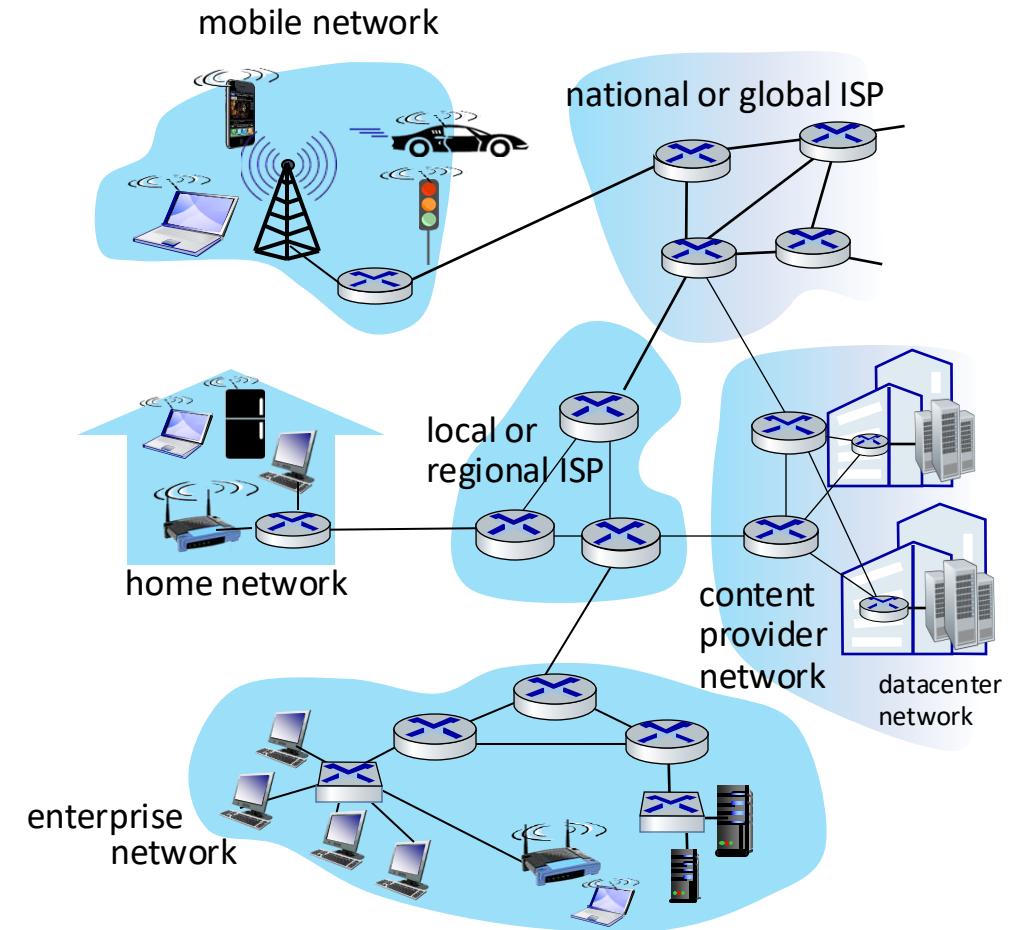
- <https://www.youtube.com/watch?v=9hIQjrMHTv4>



<https://www.internetsociety.org/internet/history-internet>

Internet structure: a “network of networks”

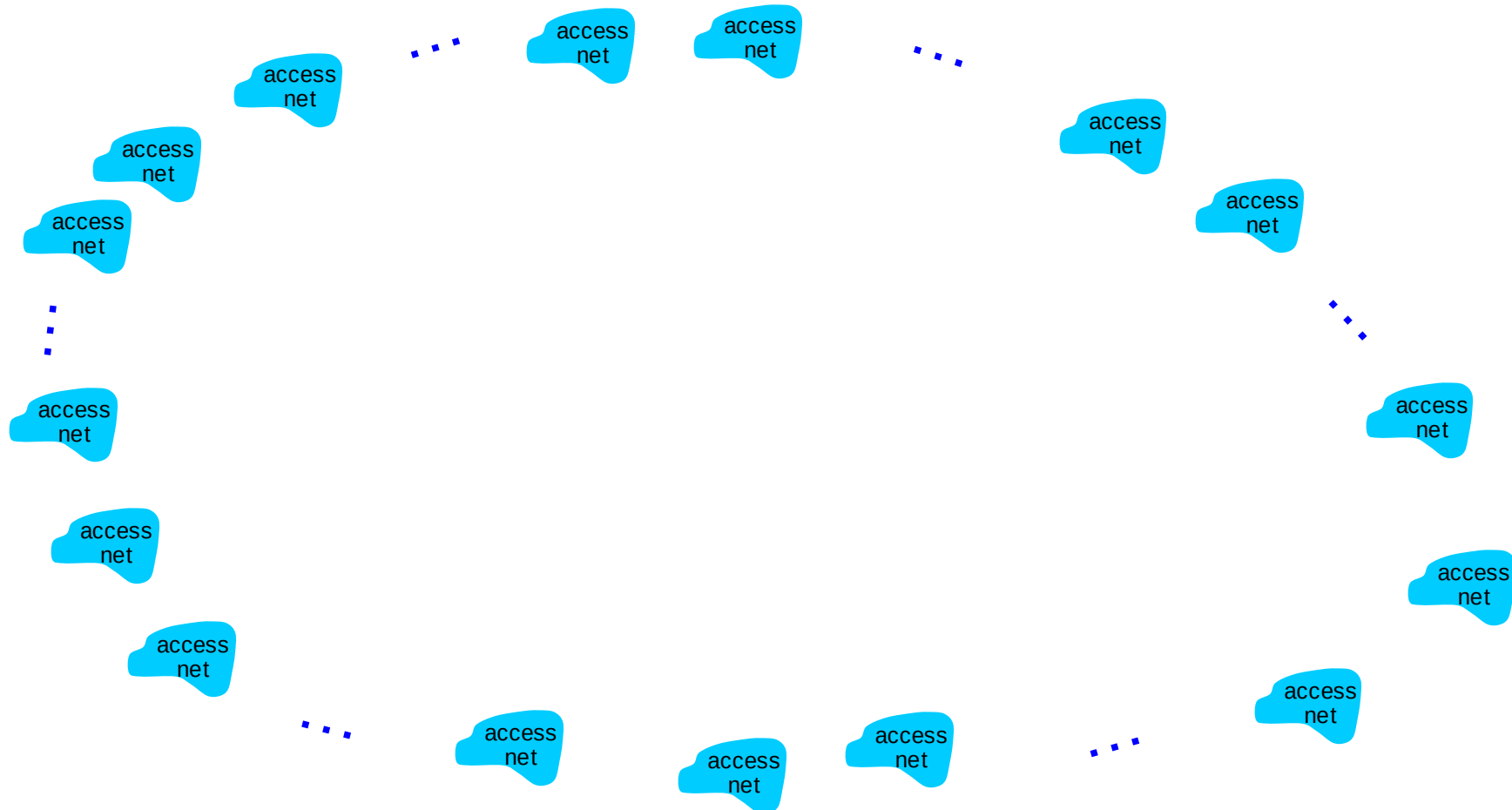
- hosts connect to Internet via **access** Internet Service Providers (ISPs)
- access ISPs in turn must be interconnected
 - so that *any* two hosts (*anywhere!*) can send packets to each other
- resulting network of networks is very complex
 - evolution driven by **economics**, **national policies**



Let's take a stepwise approach to describe current Internet structure

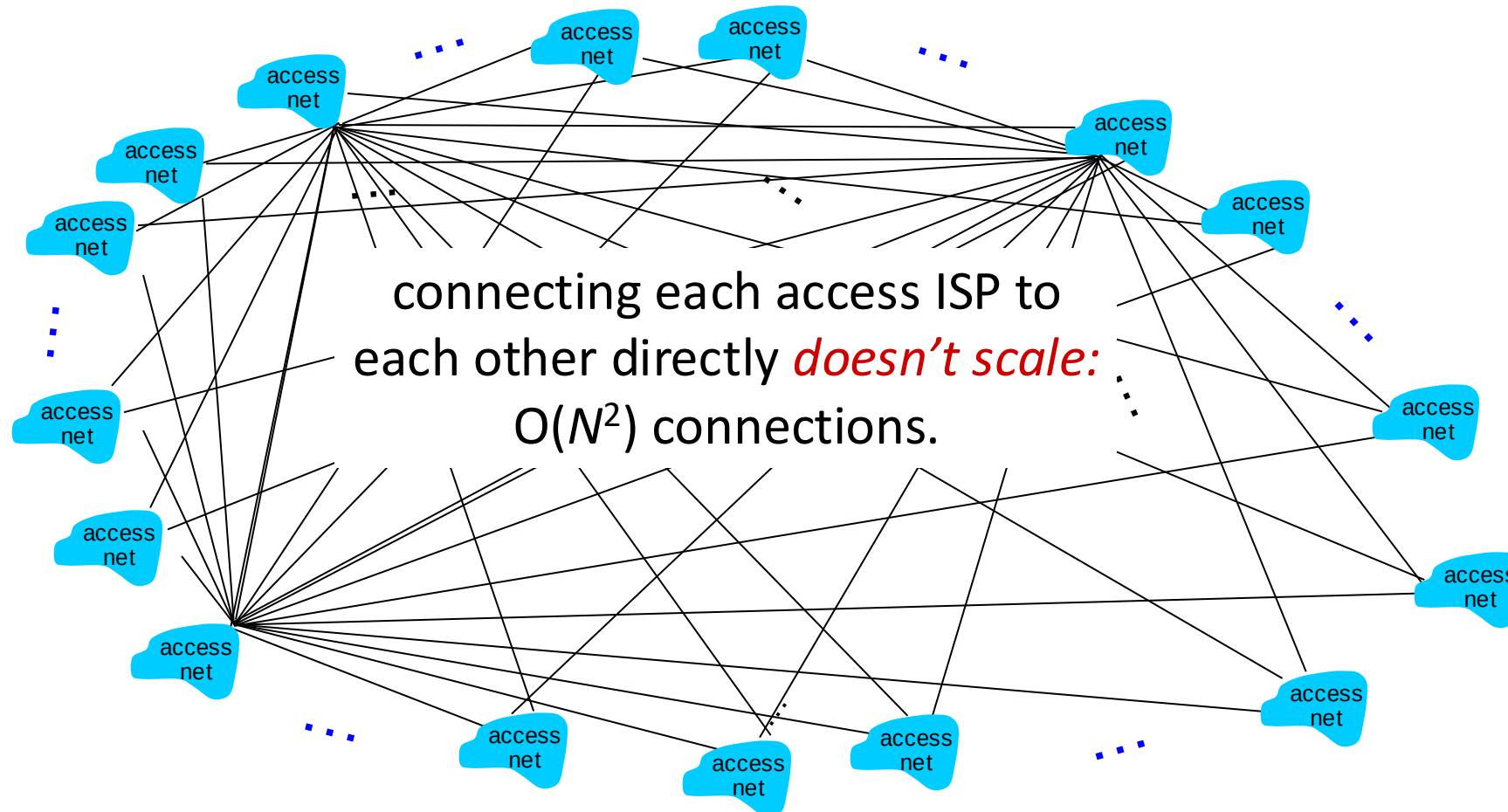
Internet structure: a “network of networks”

Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a “network of networks”

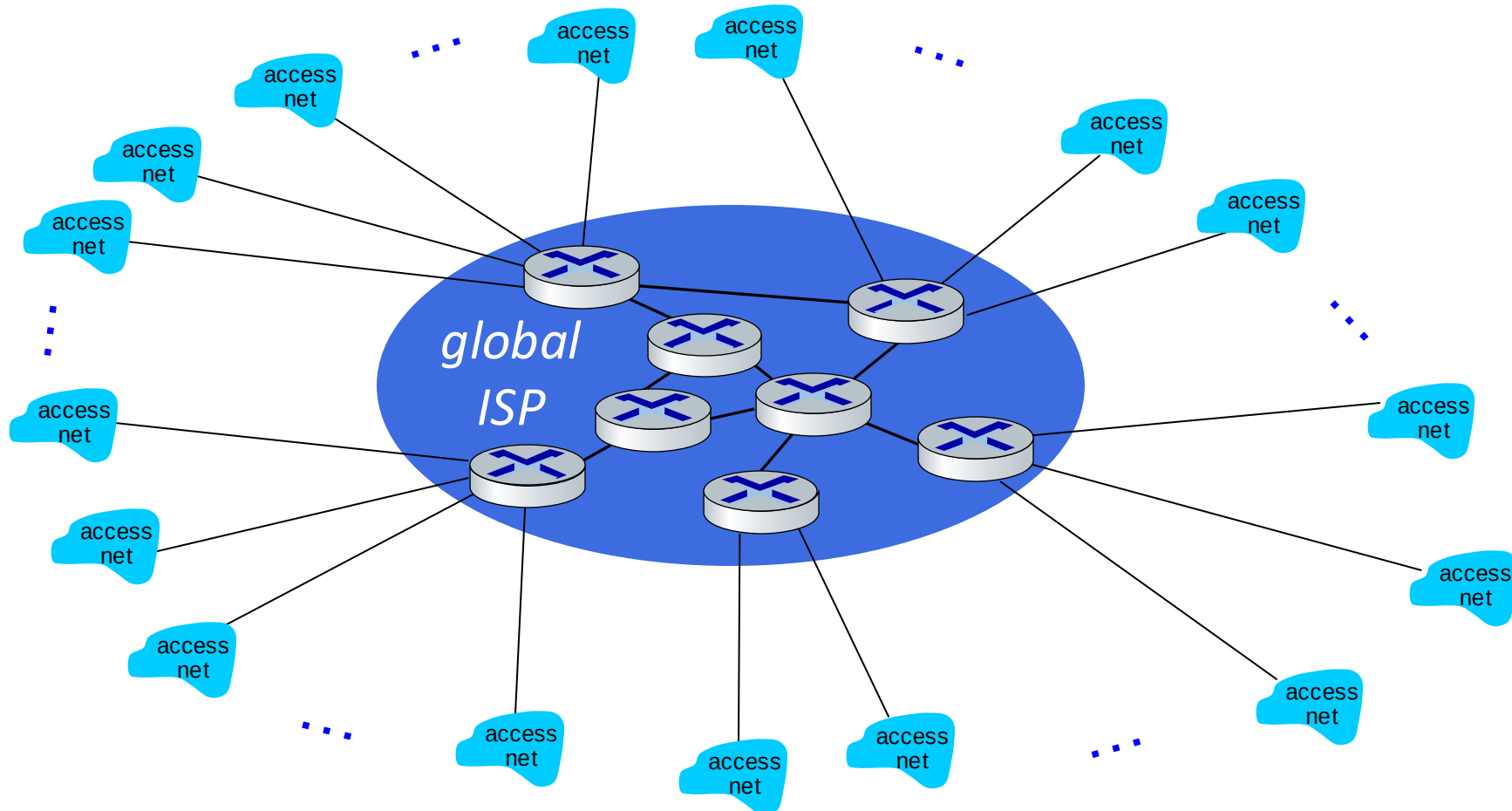
Question: given *millions* of access ISPs, how to connect them together?



Internet structure: a “network of networks”

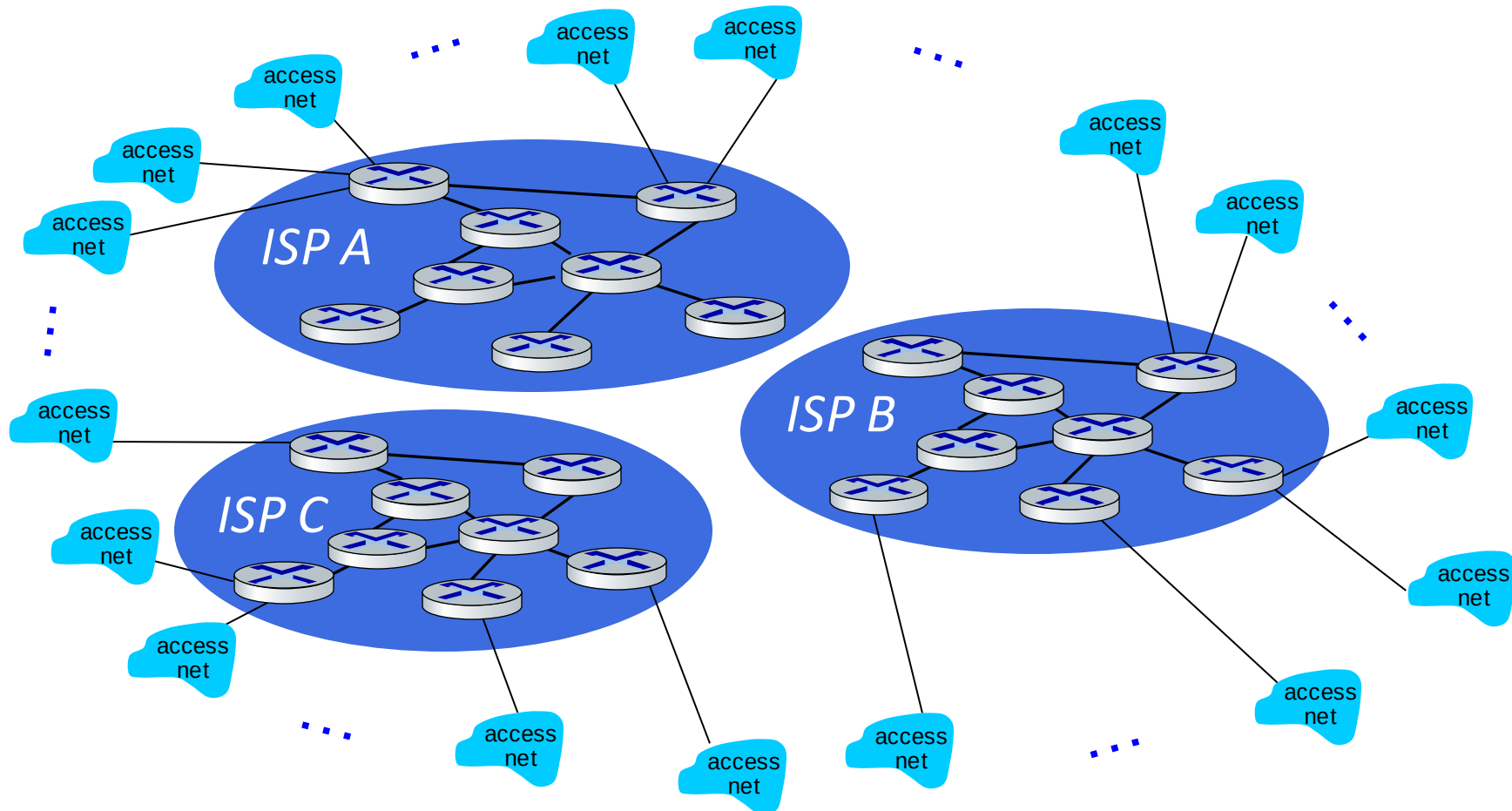
Option: connect each access ISP to one global transit ISP?

Customer and provider ISPs have economic agreement.



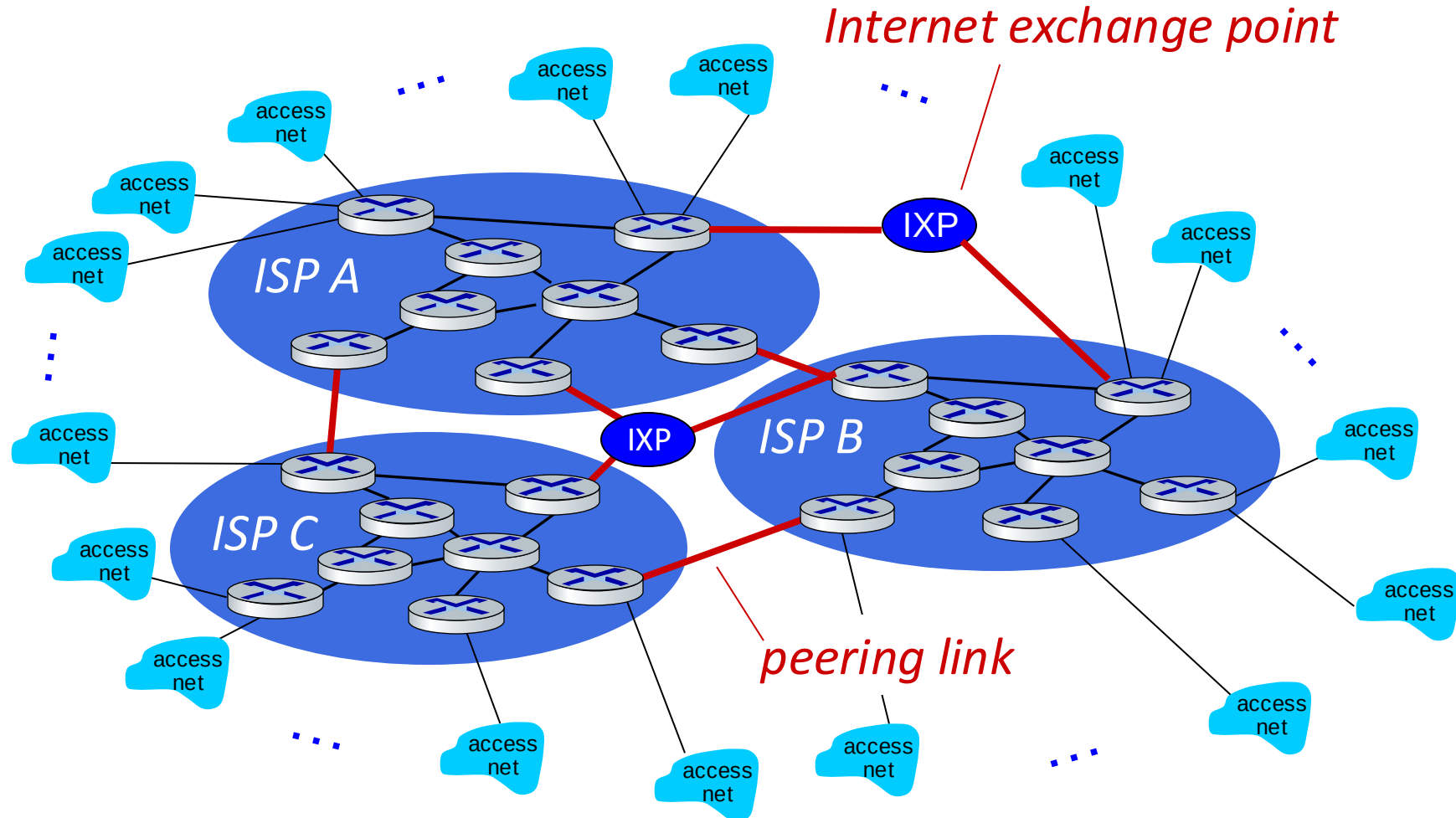
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors



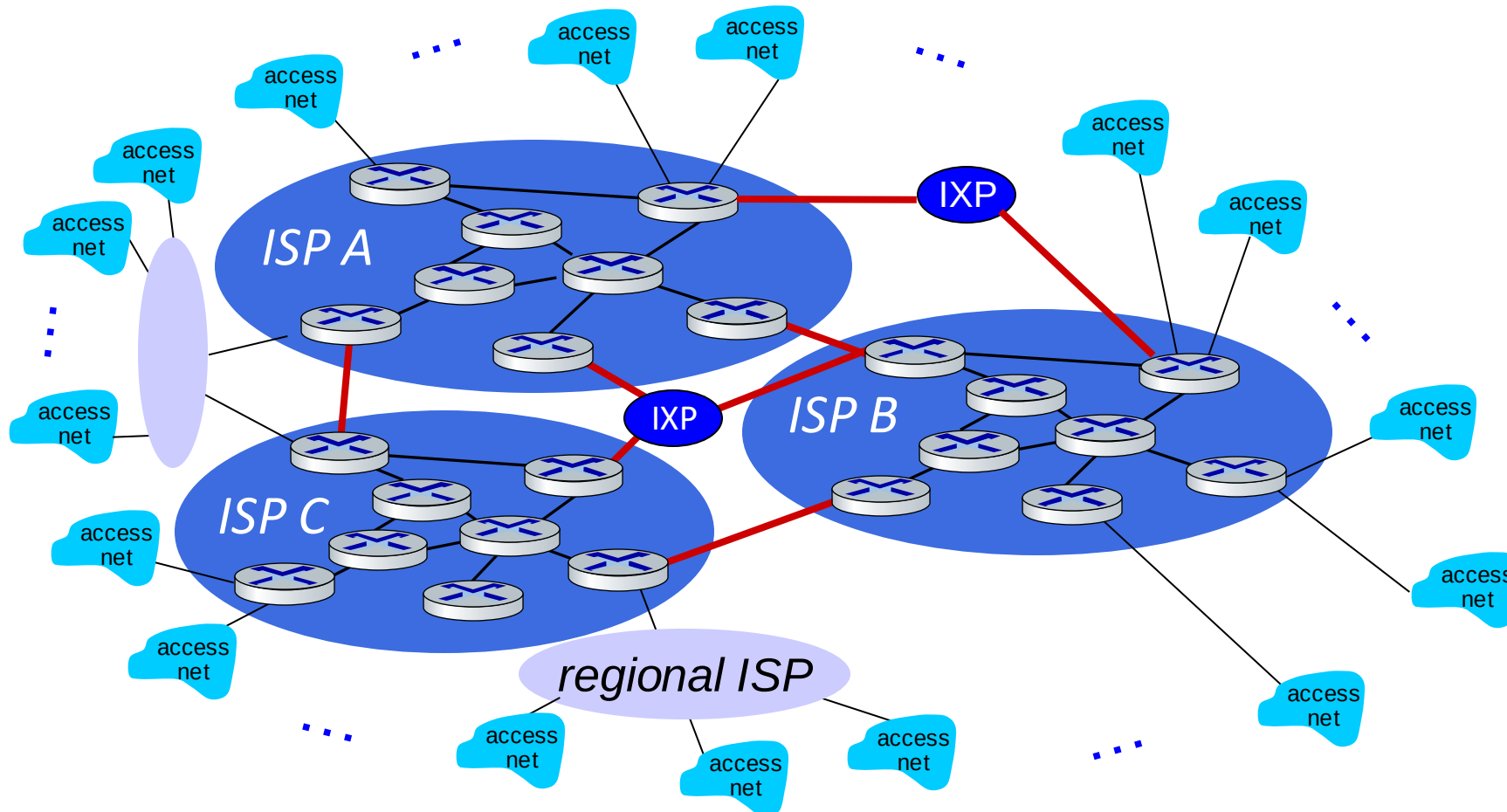
Internet structure: a “network of networks”

But if one global ISP is viable business, there will be competitors who will want to be connected



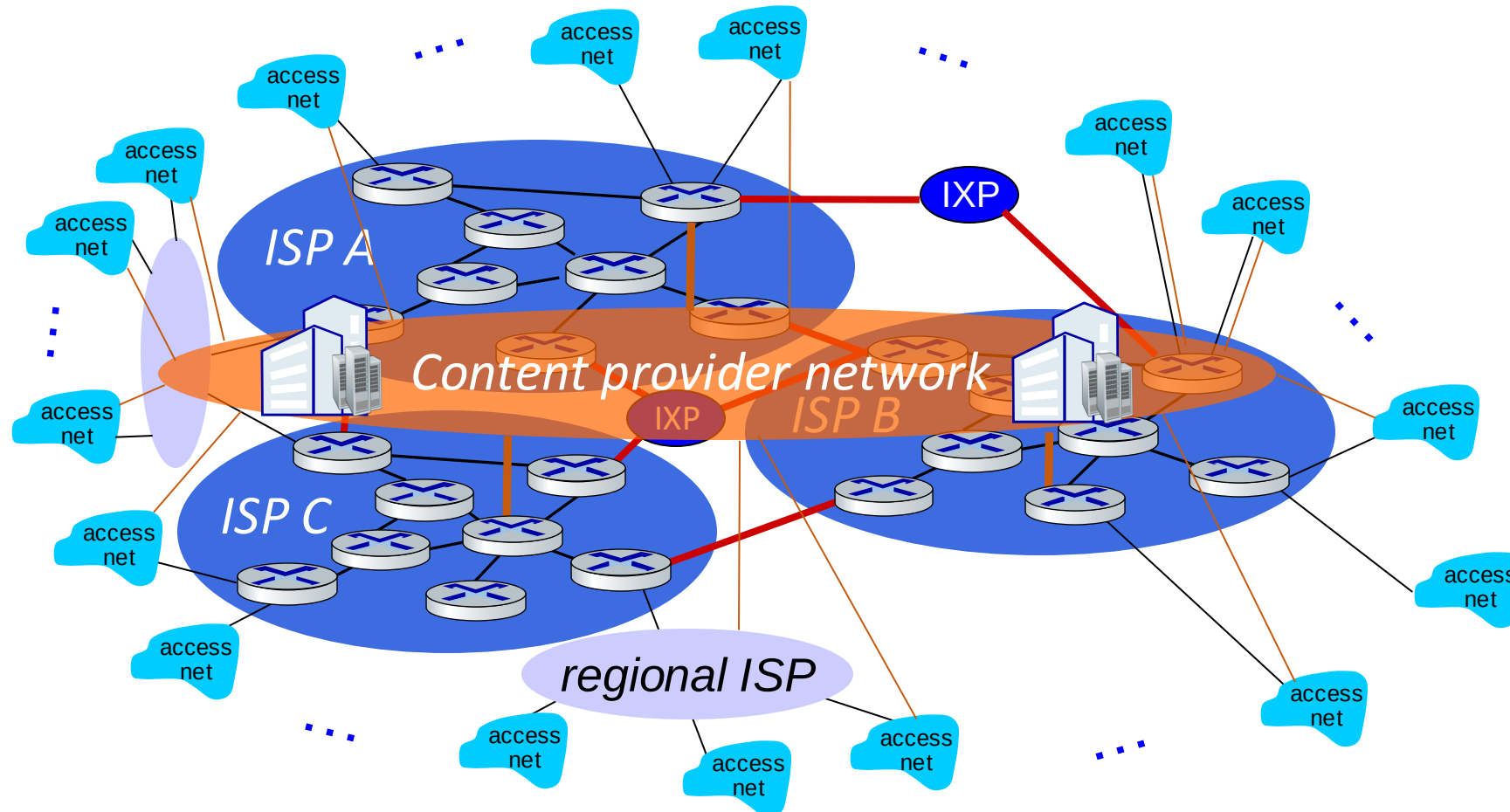
Internet structure: a “network of networks”

... and regional networks may arise to connect access nets to ISPs

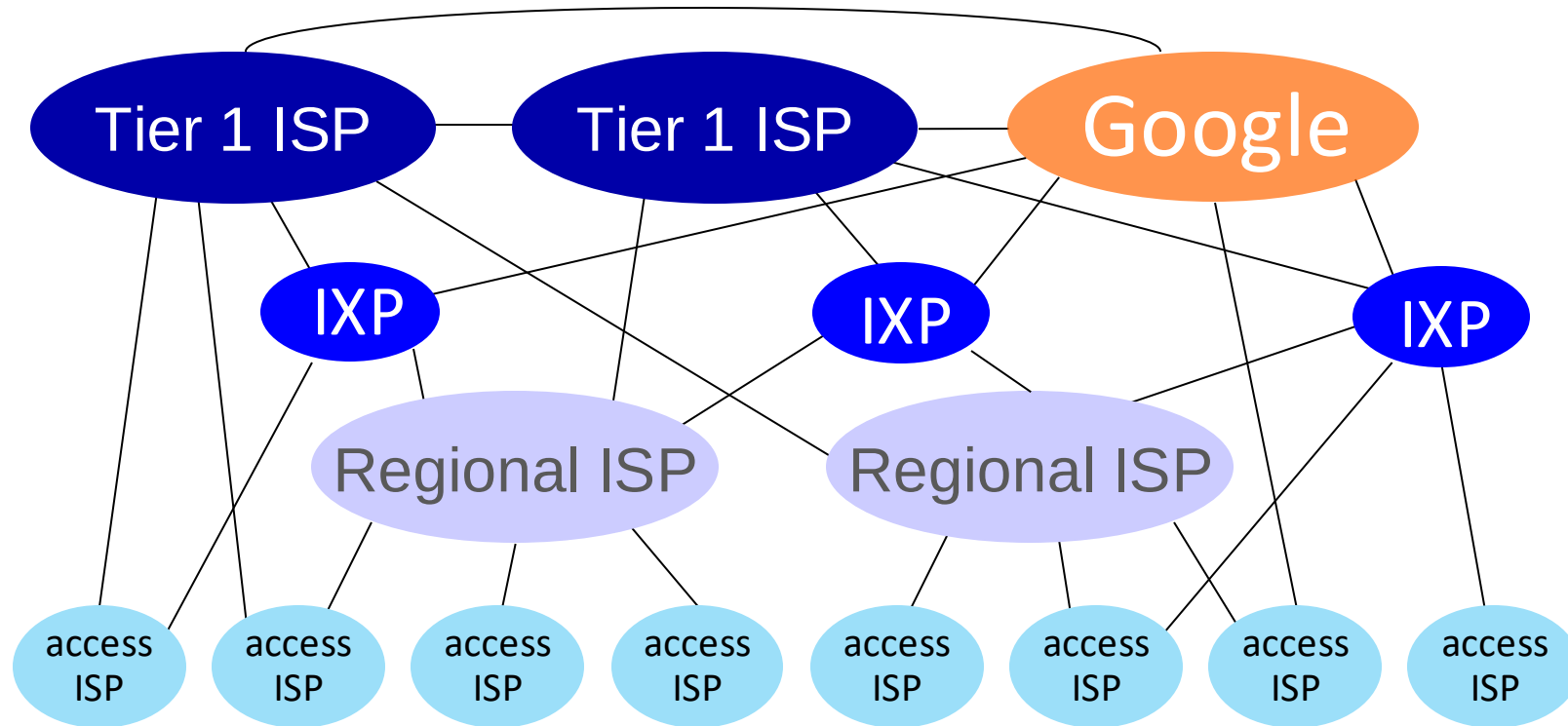


Internet structure: a “network of networks”

... and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



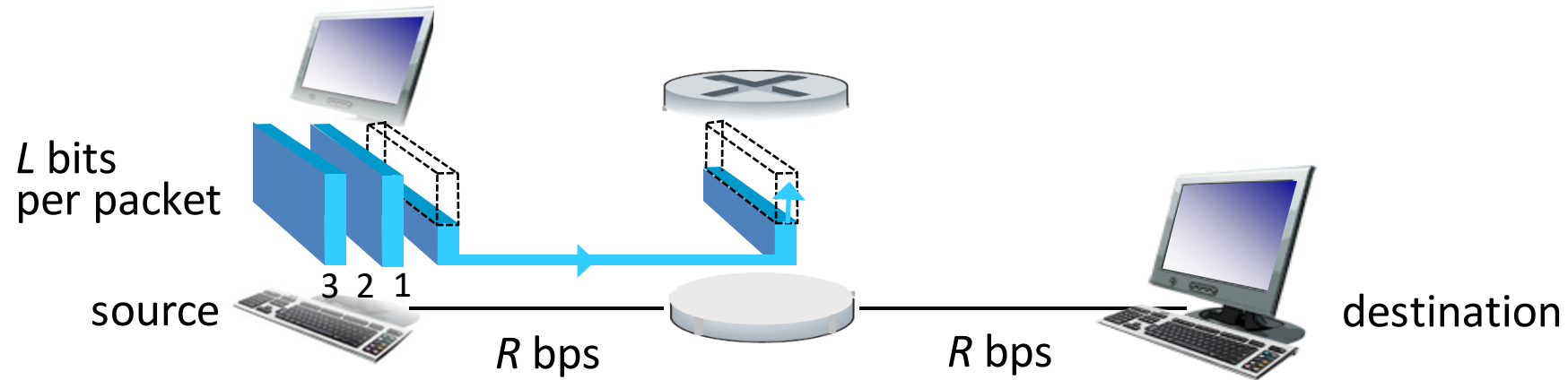
Internet structure: a “network of networks”



At “center”: small # of well-connected large networks

- **“tier-1” commercial ISPs** (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
- **content provider networks** (e.g., Google, Facebook): private network that connects its data centers to Internet, often bypassing tier-1, regional ISPs

Packet-switching: store-and-forward

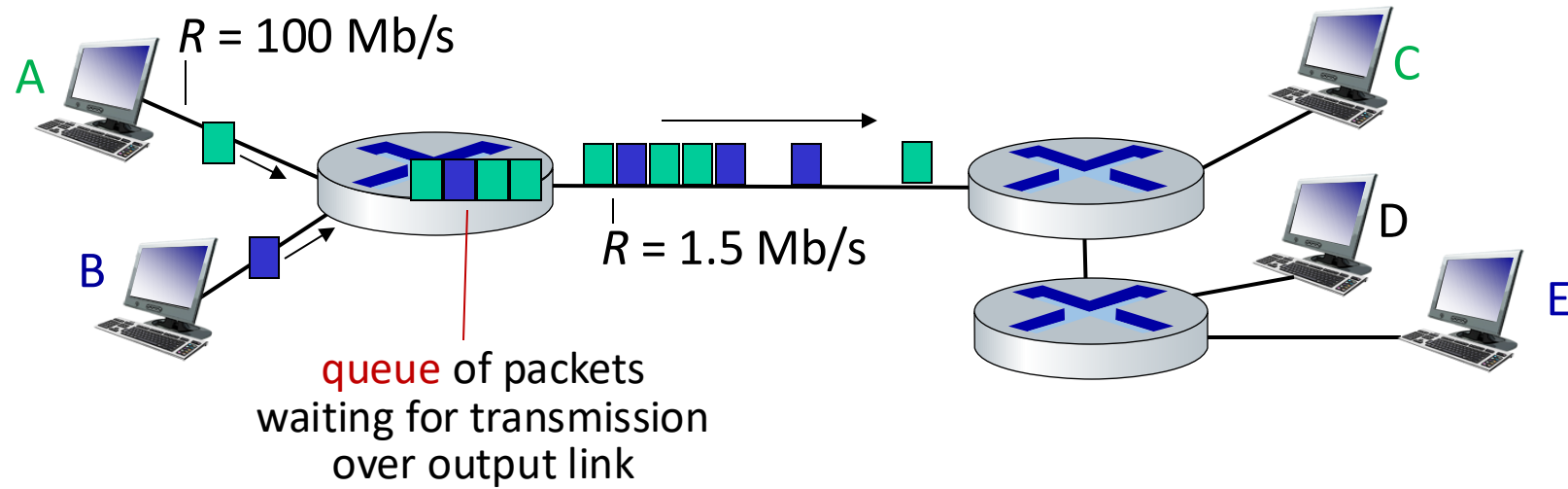


- **packet transmission delay:** takes L/R seconds to transmit (push out) L -bit packet into link at R bps
- **store and forward:** entire packet must arrive at router before it can be transmitted on next link

One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- one-hop transmission delay = 0.1 msec

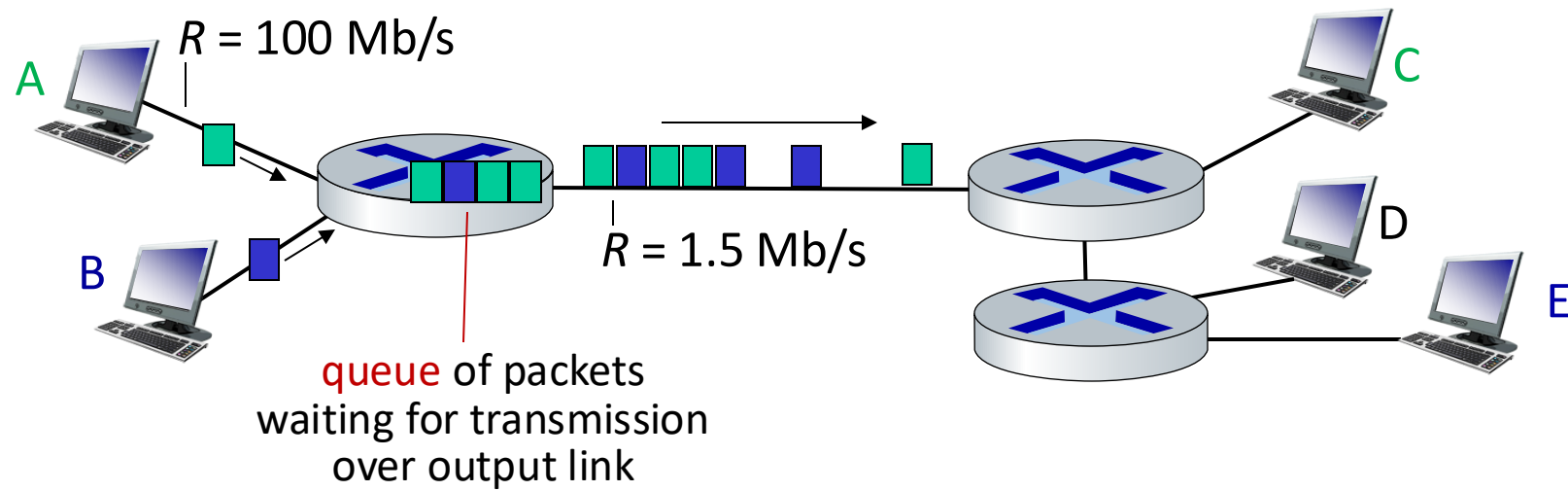
Packet-switching: queueing



Queueing occurs when work arrives faster than it can be serviced:



Packet-switching: queueing

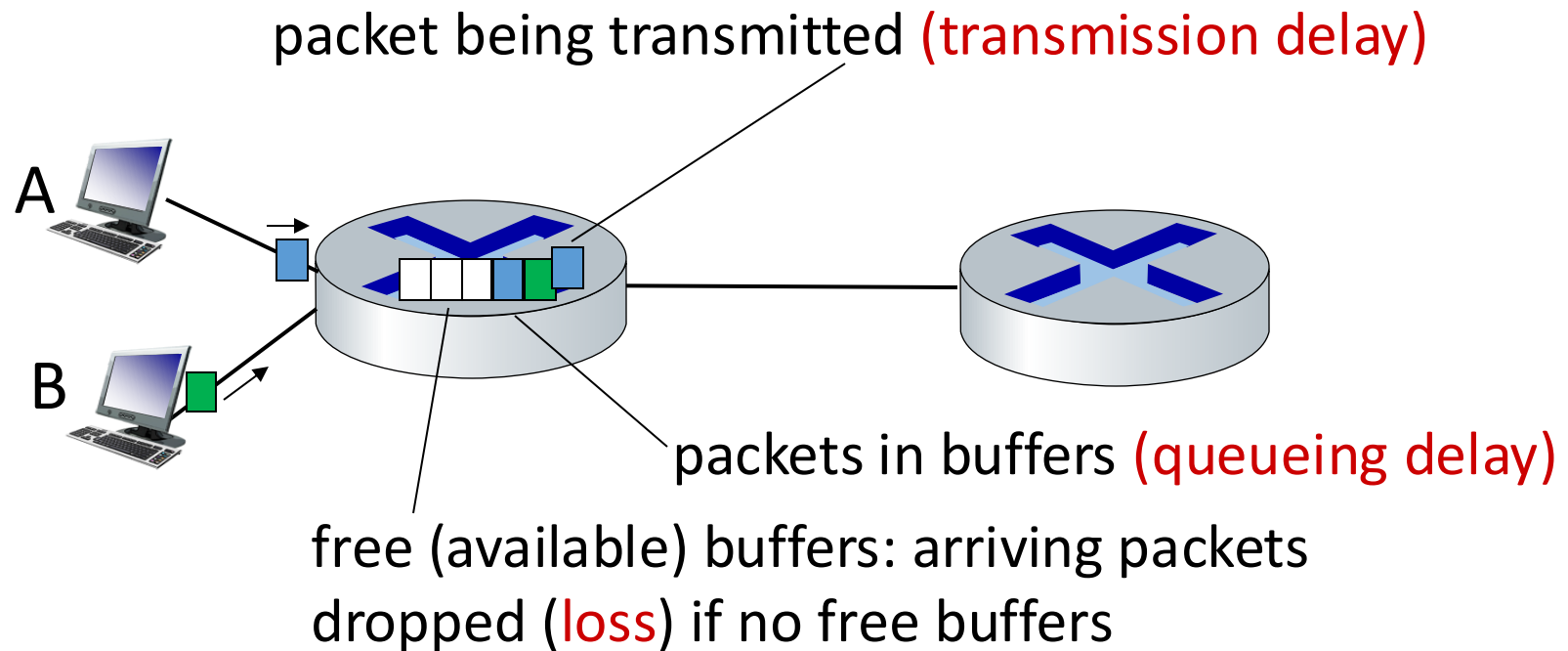


Packet queuing and loss: if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time:

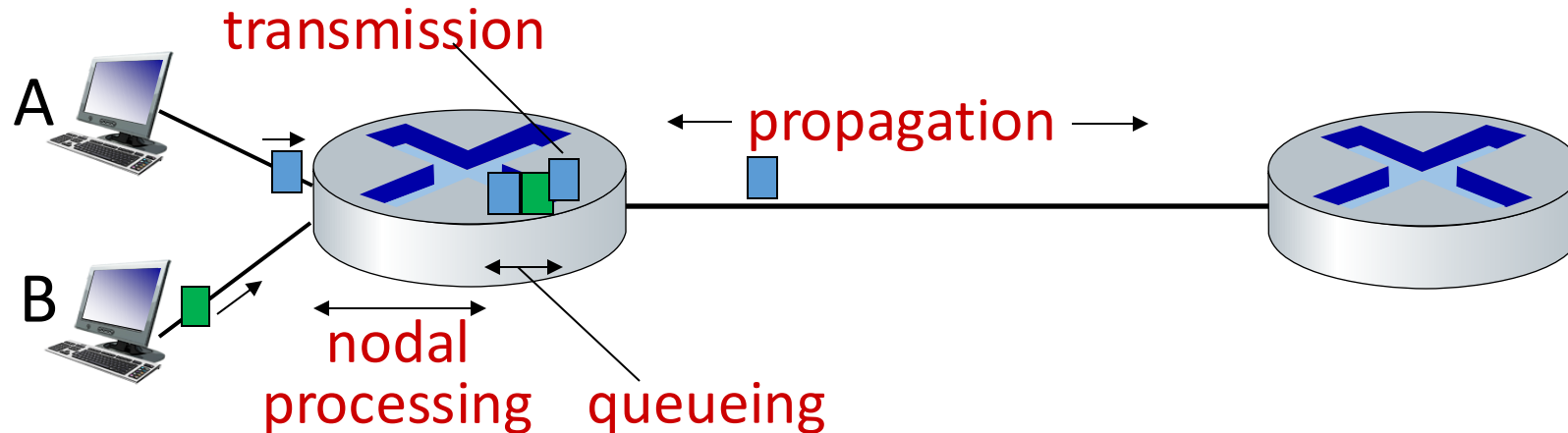
- packets will queue, waiting to be transmitted on output link
- packets can be dropped (lost) if memory (buffer) in router fills up

How do packet delay and loss occur?

- packets *queue* in router buffers, waiting for turn for transmission
 - queue length grows when arrival rate to link (temporarily) exceeds output link capacity
 - packet *loss* occurs when memory to hold queued packets fills up



Packet delay: four sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

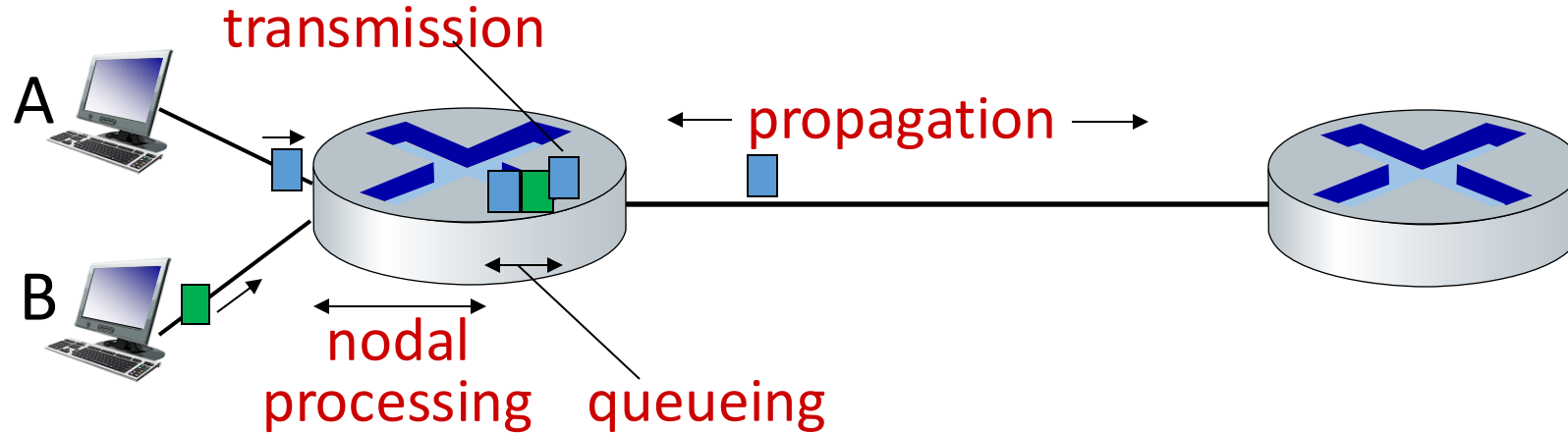
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < microsecs

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Packet delay: four sources



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link *transmission rate* (bps)

■ $d_{\text{trans}} = L/R$

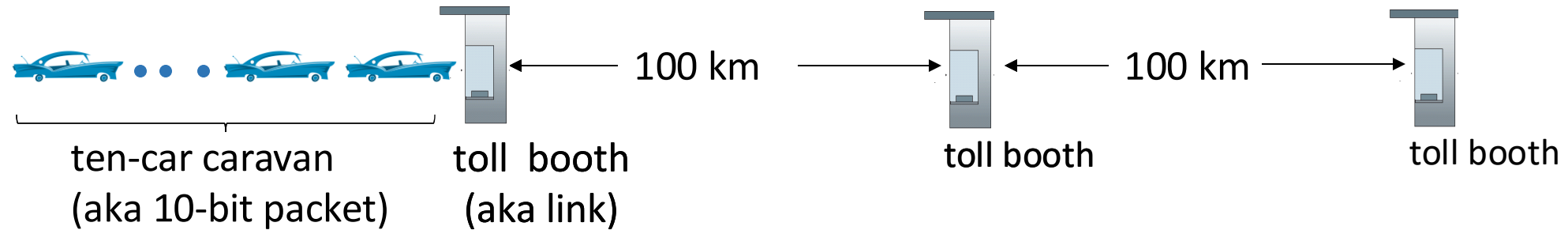
d_{prop} : propagation delay:

- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

■ $d_{\text{prop}} = d/s$

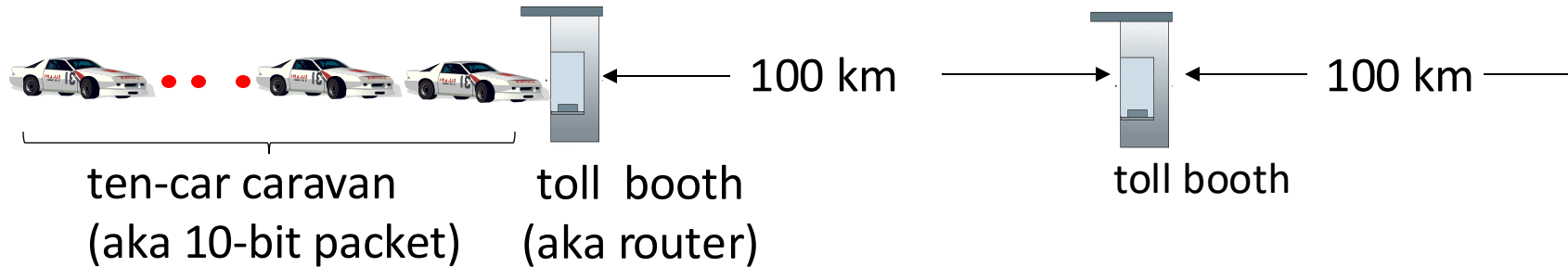
d_{trans} and d_{prop}
very different

Caravan analogy



- car $\sim$ bit; caravan $\sim$ packet; toll service $\sim$ link transmission
- toll booth takes 12 sec to service car (bit transmission time)
- “propagate” at 100 km/hr
- **Q: How long until caravan is lined up before 2nd toll booth?**
- time to “push” entire caravan through toll booth onto highway = $12 * 10 = 120$ sec
- time for last car to propagate from 1st to 2nd toll booth: $100\text{km} / (100\text{km/hr}) = 1$ hr
- **A: 62 minutes**

Caravan analogy



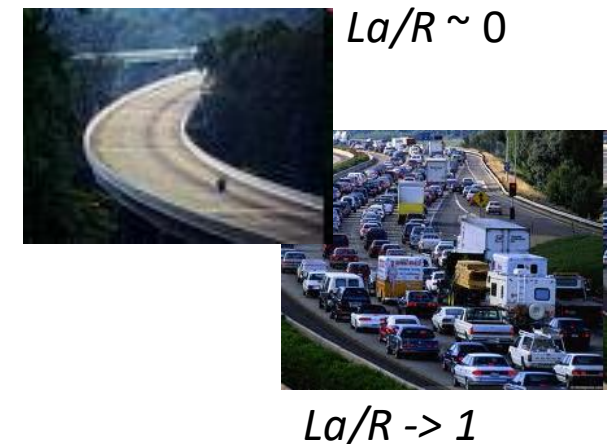
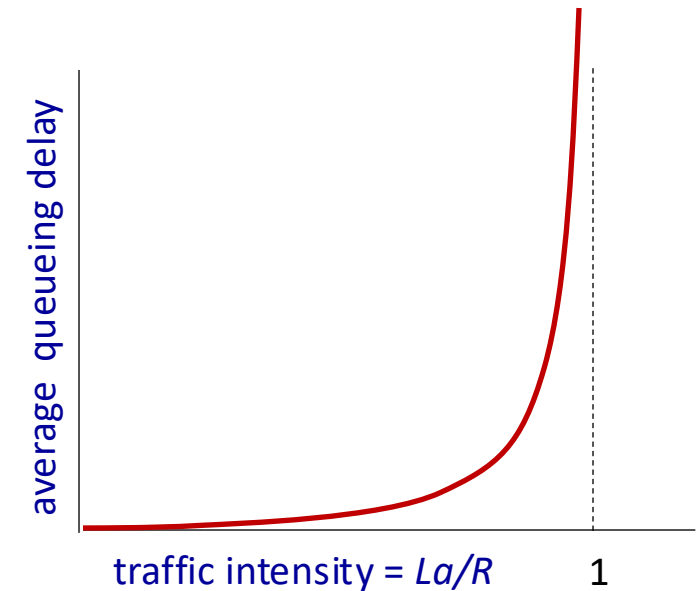
- suppose cars now “propagate” at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- **Q: Will cars arrive to 2nd booth before all cars serviced at first booth?**
A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

Packet queueing delay (revisited)

- a : average packet arrival rate
- L : packet length (bits)
- R : link bandwidth (bit transmission rate)

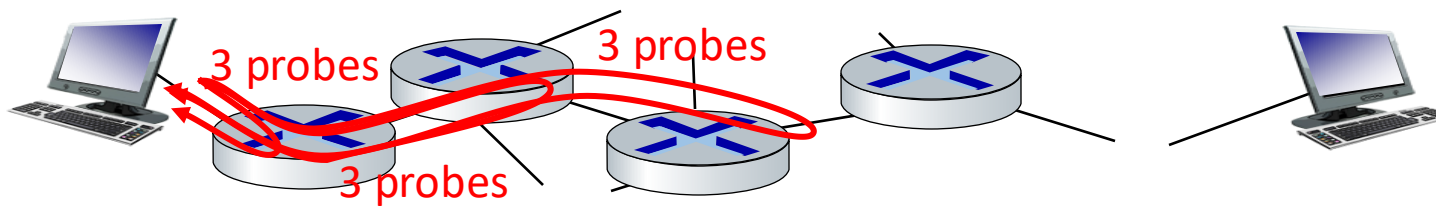
$$\frac{L \cdot a}{R} : \frac{\text{arrival rate of bits}}{\text{service rate of bits}} \quad \text{“traffic intensity”}$$

- $La/R \sim 0$: avg. queueing delay small
- $La/R \rightarrow 1$: avg. queueing delay large
- $La/R > 1$: more “work” arriving is more than can be serviced - average delay infinite!



“Real” Internet delays and routes

- what do “real” Internet delay & loss look like?
- **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination. For all i :
 - sends three packets that will reach router i on path towards destination (with time-to-live field value of i)
 - router i will return packets to sender
 - sender measures time interval between transmission and reply



Real Internet delays and routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu

3 delay measurements
to border1-rt-fa5-1-0.gw.umass.edu

trans-oceanic link

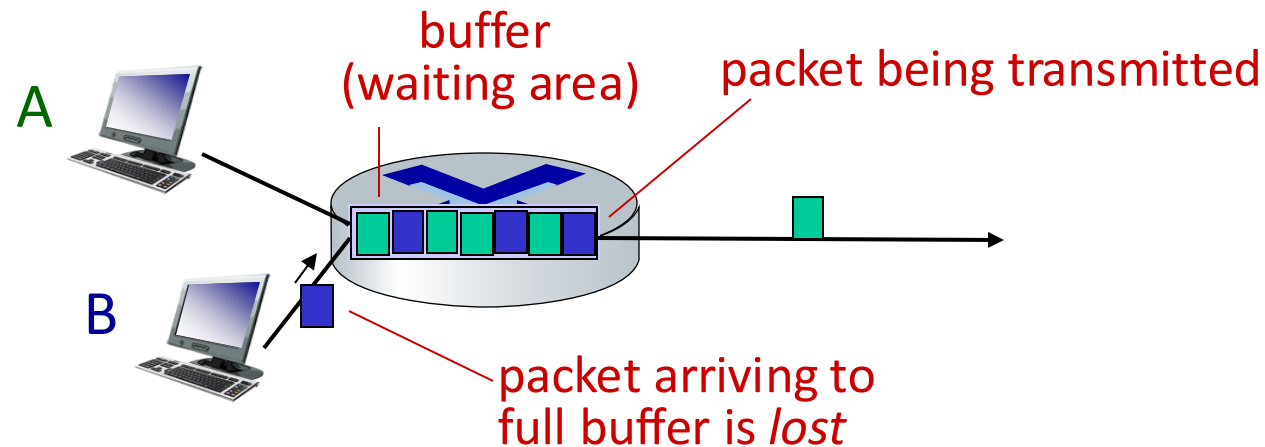
looks like delays
decrease! Why?

* means no response (probe lost, router not replying)

1	cs-gw (128.119.240.254)	1 ms	1 ms	2 ms
2	border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)	1 ms	1 ms	2 ms
3	cht-vbns.gw.umass.edu (128.119.3.130)	6 ms	5 ms	5 ms
4	jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)	16 ms	11 ms	13 ms
5	jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)	21 ms	18 ms	18 ms
6	abilene-vbns.abilene.ucaid.edu (198.32.11.9)	22 ms	18 ms	22 ms
7	nycm-wash.abilene.ucaid.edu (198.32.8.46)	22 ms	22 ms	22 ms
8	62.40.103.253 (62.40.103.253)	104 ms	109 ms	106 ms
9	de2-1.de1.de.geant.net (62.40.96.129)	109 ms	102 ms	104 ms
10	de.fr1.fr.geant.net (62.40.96.50)	113 ms	121 ms	114 ms
11	renater-gw.fr1.fr.geant.net (62.40.103.54)	112 ms	114 ms	112 ms
12	nio-n2.cssi.renater.fr (193.51.206.13)	111 ms	114 ms	116 ms
13	nice.cssi.renater.fr (195.220.98.102)	123 ms	125 ms	124 ms
14	r3t2-nice.cssi.renater.fr (195.220.98.110)	126 ms	126 ms	124 ms
15	eurecom-valbonne.r3t2.ft.net (193.48.50.54)	135 ms	128 ms	133 ms
16	194.214.211.25 (194.214.211.25)	126 ms	128 ms	126 ms
17	***			
18	***			
19	fantasia.eurecom.fr (193.55.113.142)	132 ms	128 ms	136 ms

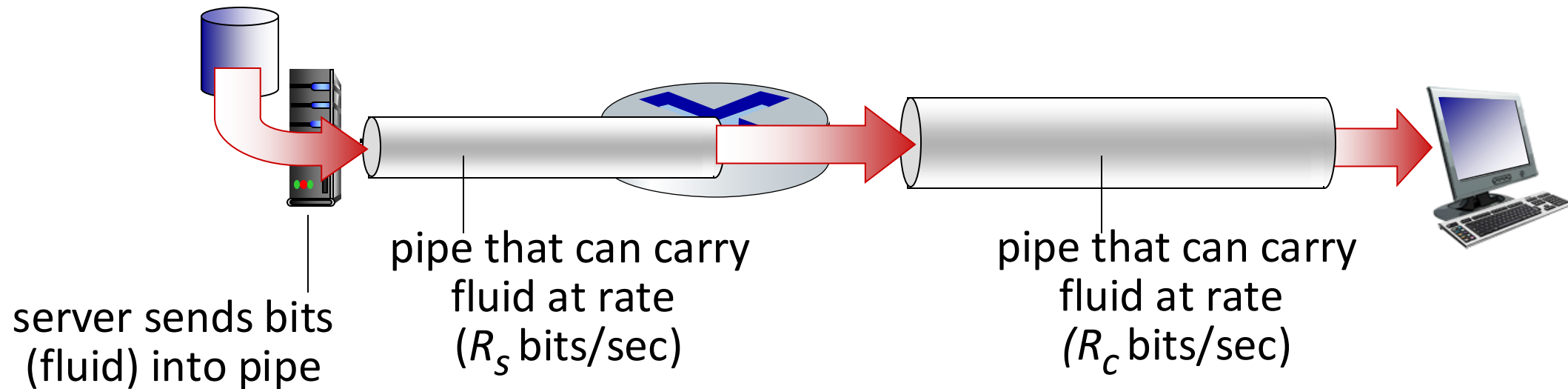
Packet loss

- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet may be retransmitted by previous node, by source end system, or not at all



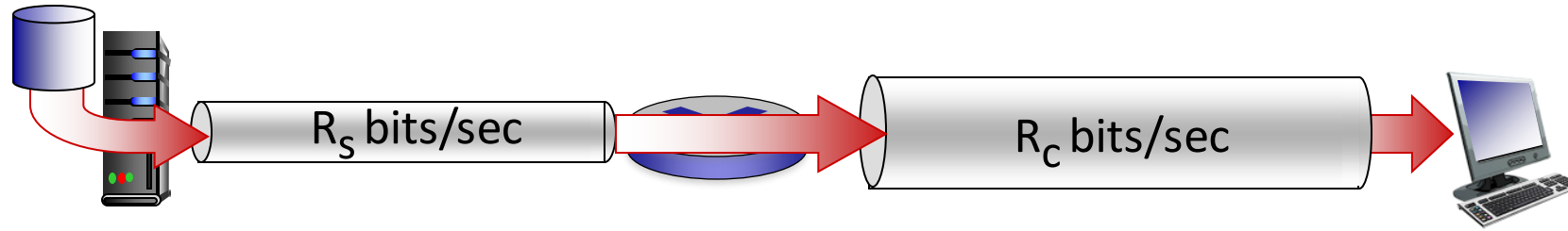
Throughput

- *throughput*: rate (bits/time unit) at which bits are being sent from sender to receiver
 - *instantaneous*: rate at given point in time
 - *average*: rate over longer period of time

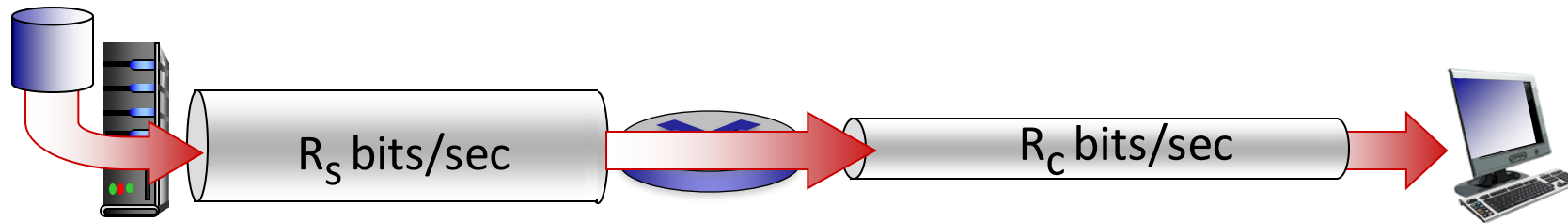


Throughput

$R_s < R_c$ What is average end-end throughput?



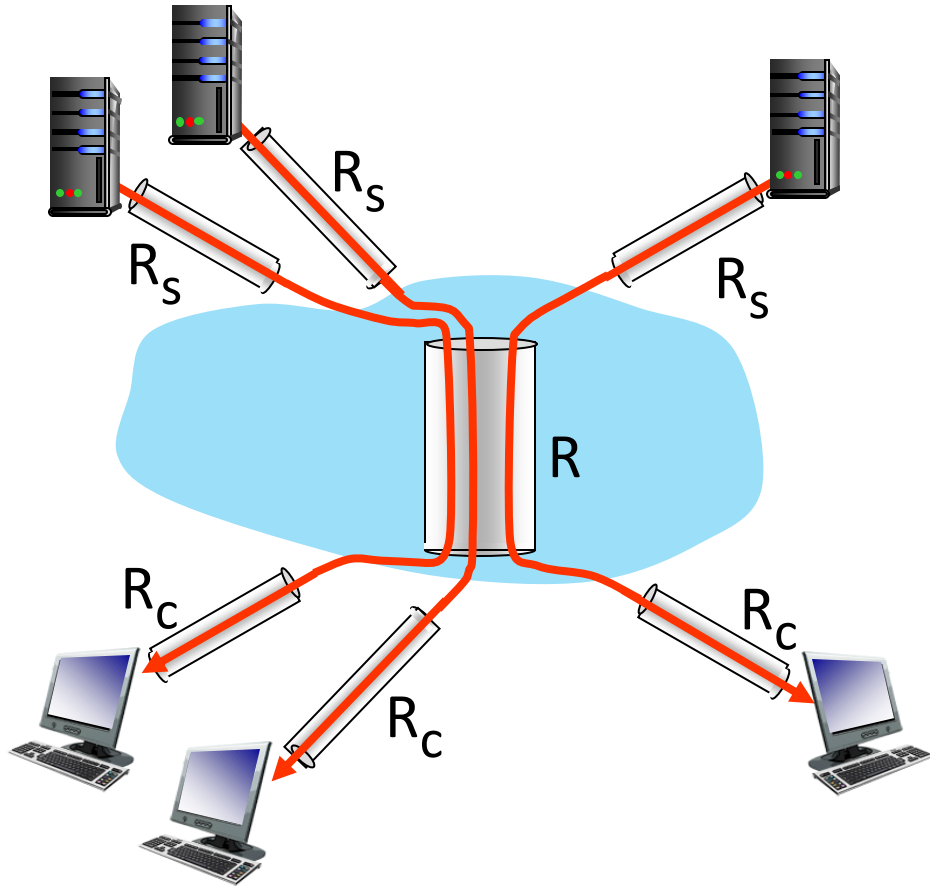
$R_s > R_c$ What is average end-end throughput?



bottleneck link

link on end-end path that constrains end-end throughput

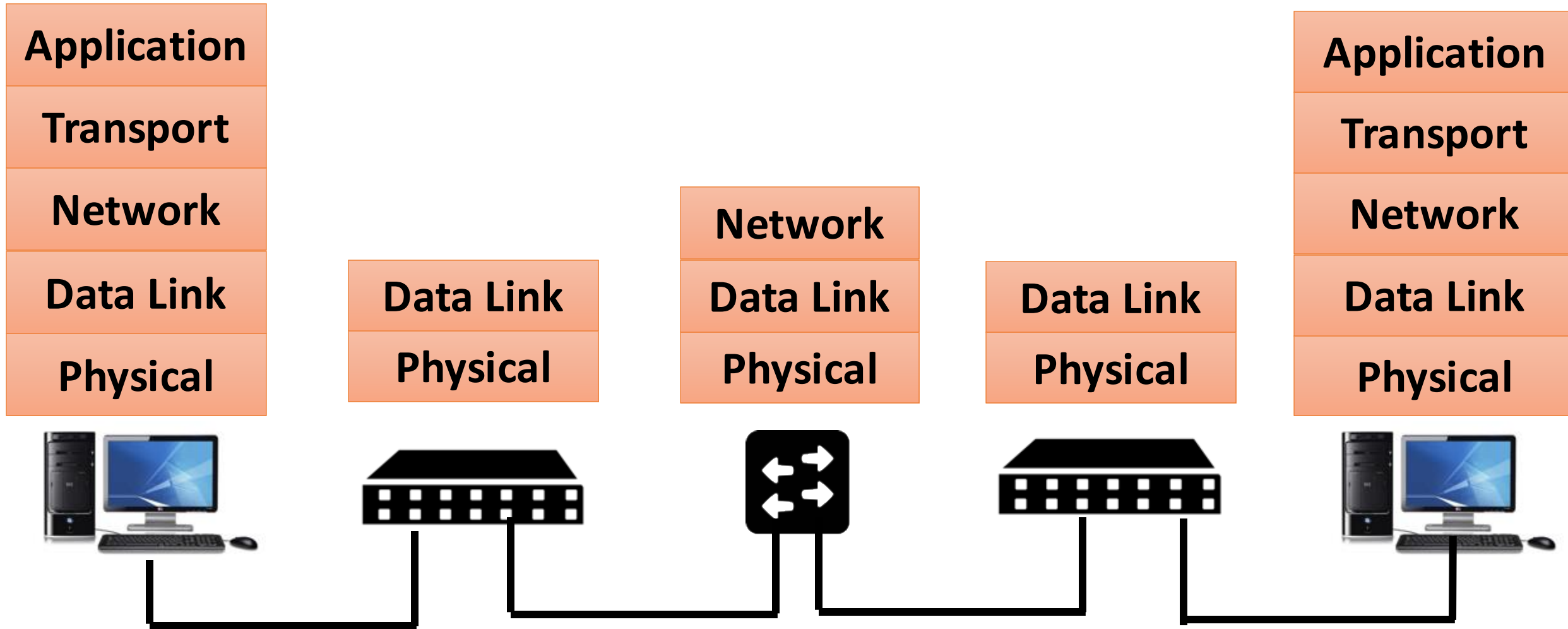
Throughput: network scenario



10 connections (fairly) share
backbone bottleneck link R bits/sec

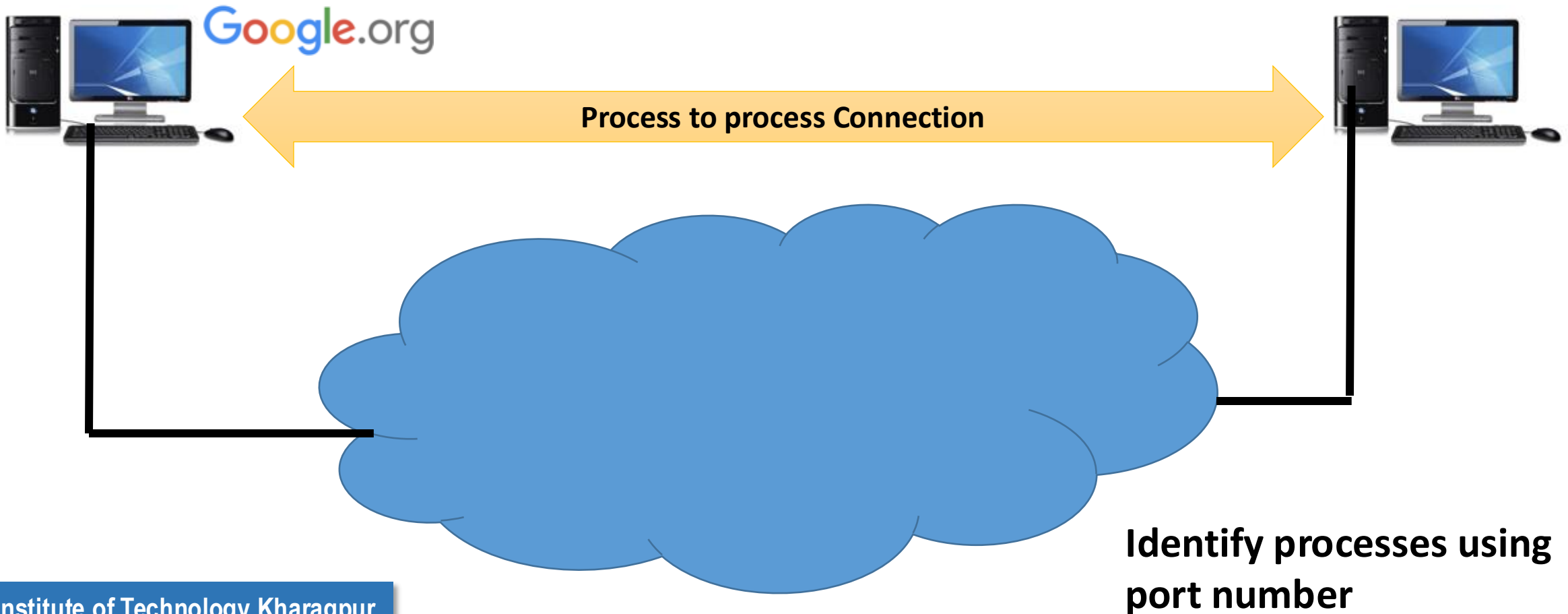
- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck

TCP/IP Protocol Stack



Transport Layer Services

- Connection oriented communication
 - Create a logical connection between your machine and Google server



Transport Layer Services

- Connection oriented communication
 - Create a logical connection between your machine and Google server
- Ordered delivery of data packets
- Reliability
- Flow control
- Congestion avoidance

Network (Internet) Layer Services

- Connectionless communication - construct messages (datagrams) from packets and route them to the next hop
 - Why do transport layer packets need to be converted to datagrams at every hop?
- Host addressing - use IP addresses to uniquely identify a host in the network
- Datagram routing - decide the path to route every datagram from the network graph

Data Link Layer Services

- Encapsulation of network layer data packets into frames
- Frame synchronization
- Error control
- Flow control
 - Why do we need flow control at the link layer?
- Channel access / medium access, Physical addressing

Data Link Layer

Logical Link Control (LLC)

Error control and flow control

Medium Access Control (MAC)

Channel access and physical layer addressing

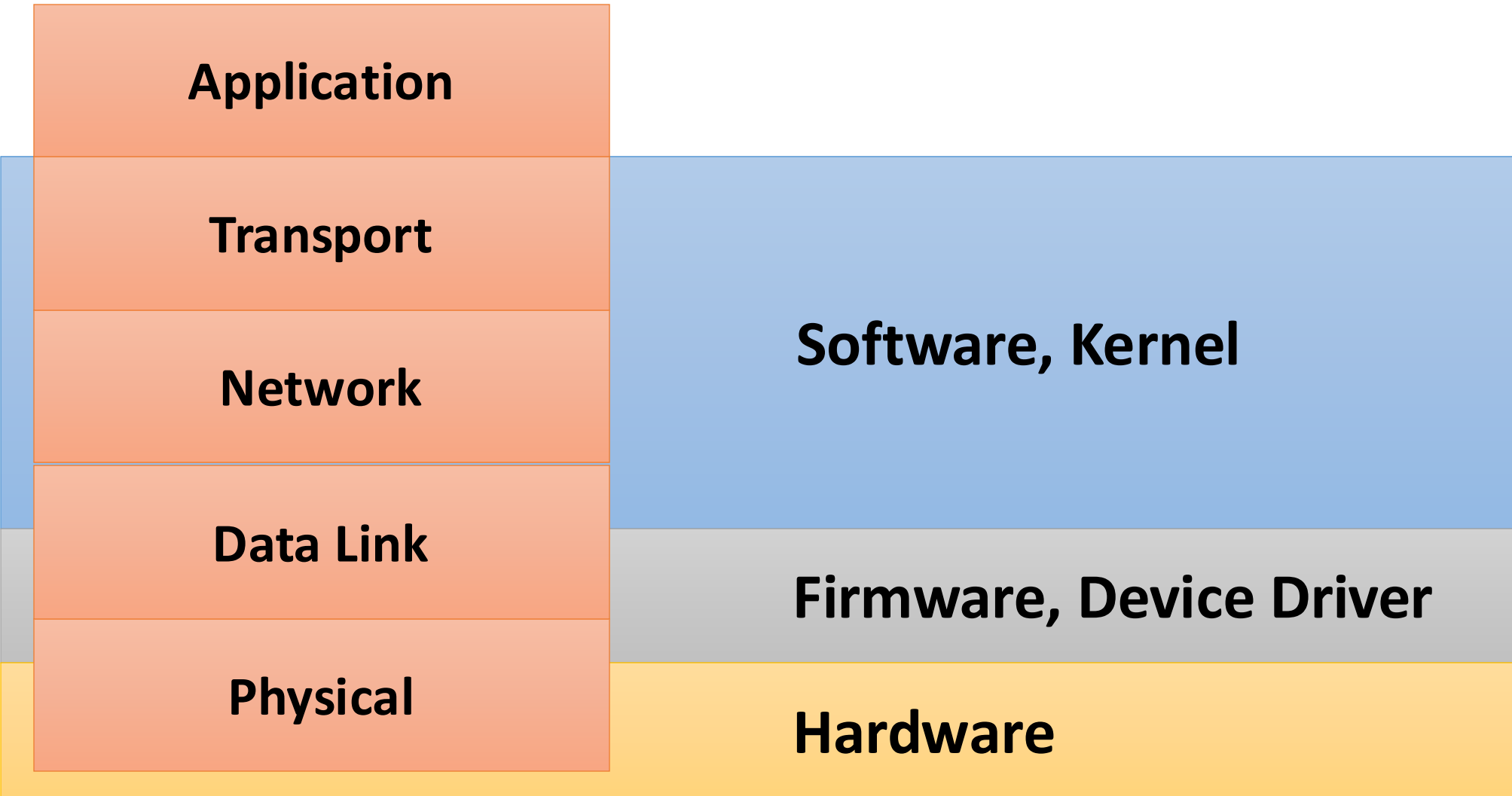
Addressing a Host in the Network

- Every host has two addresses –
 - Logical address (IP address) – used to find out the path towards a host
 - Physical address (MAC address) - uniquely identify a host in the Internet
- Logical address – Dr. Sandip Chakraborty, Room-311, Department of Computer Science and Engineering, IIT Kharagpur, Kharagpur - 721302, West Bengal, India
- Physical address – My Aadhar number (XXXX XXXX XXXX)

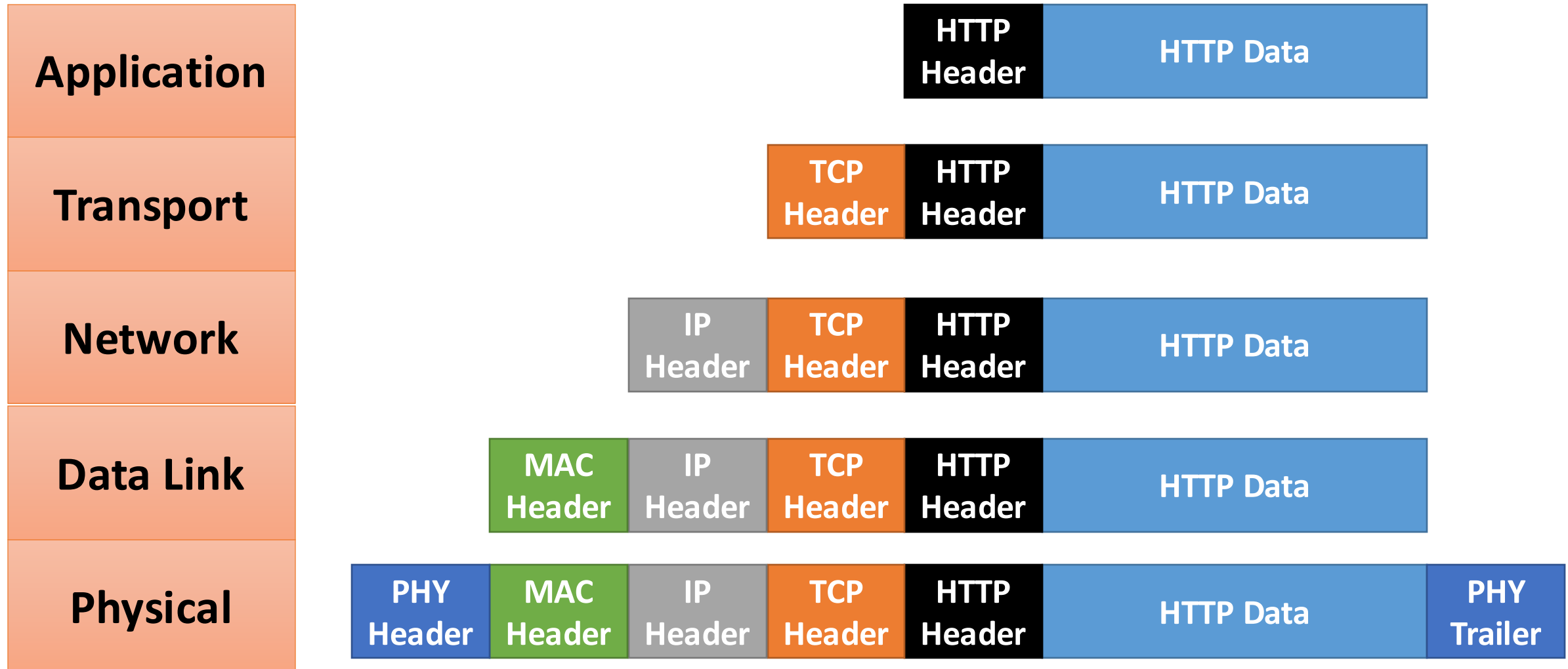
Addressing a Host in the Network

- Every host has two addresses –
 - Logical address (IP address) – used to find out the path towards a host
 - Physical address (MAC address) - uniquely identify a host in the Internet
- Logical address (32 bit IP address) – 172.16.32.64
- Physical address (64 bit MAC address) – 00:00:00:14:22:01:2C:4D

Protocol Stack Implementation in a Host

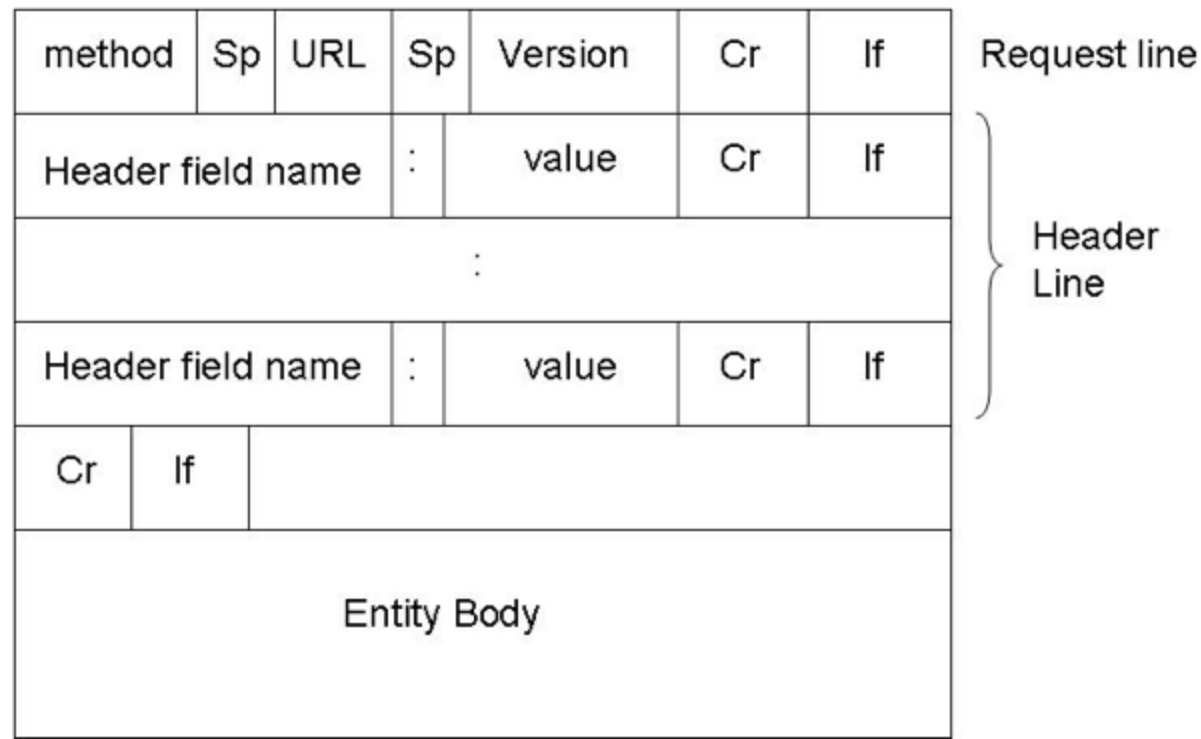


How Application Data Passes Through Different Layers



How do you access a page at www.google.com?

1. **Application Layer:** Use DNS to get the IP address of the Google server - DNS returns 74.125.224.72
2. **Application Layer:** Construct an HTTP GET Request –
GET 74.125.224.72/index.html HTTP/1.1
Construct an HTTP packet and forward it to the transport layer



How do you access a page at www.google.com?

3. Transport Layer: Construct the transport layer packet (TCP packet)

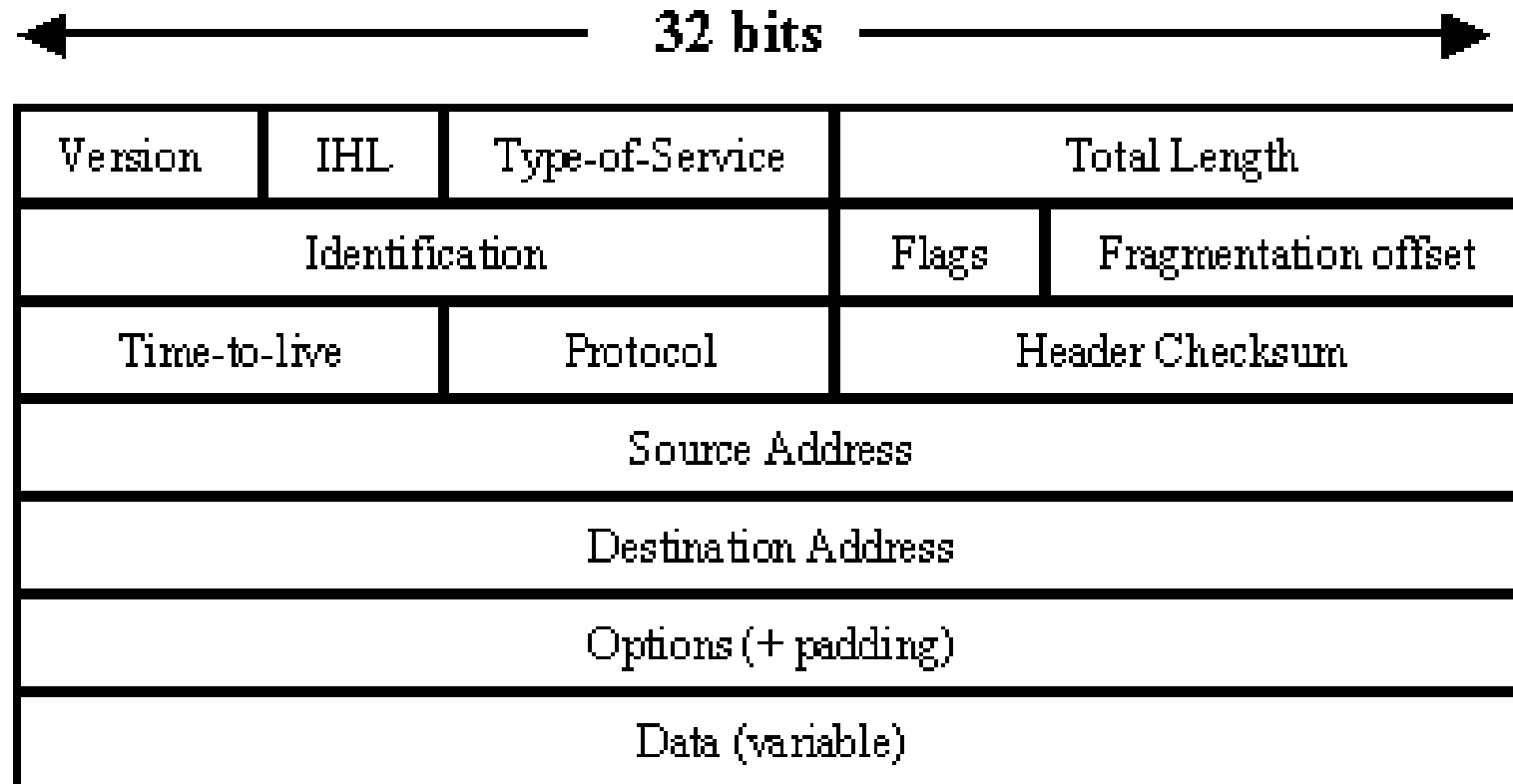
- Source port: 3324 (port address corresponding to your browser tab)
- Destination port: 80 (port for a HTTP server)

	3	9	15	23	31										
1	Source Port						Destination Port								
2	Sequence Number														
3	Acknowledgment Number														
4	Data Offset	Reserved	U R G	A C K	P S H	R S T	S Y N	F I N	Window						
5	Checksum						Urgent Pointer								
6	Options									Padding					
	Data begins here..														

How do you access a page at www.google.com?

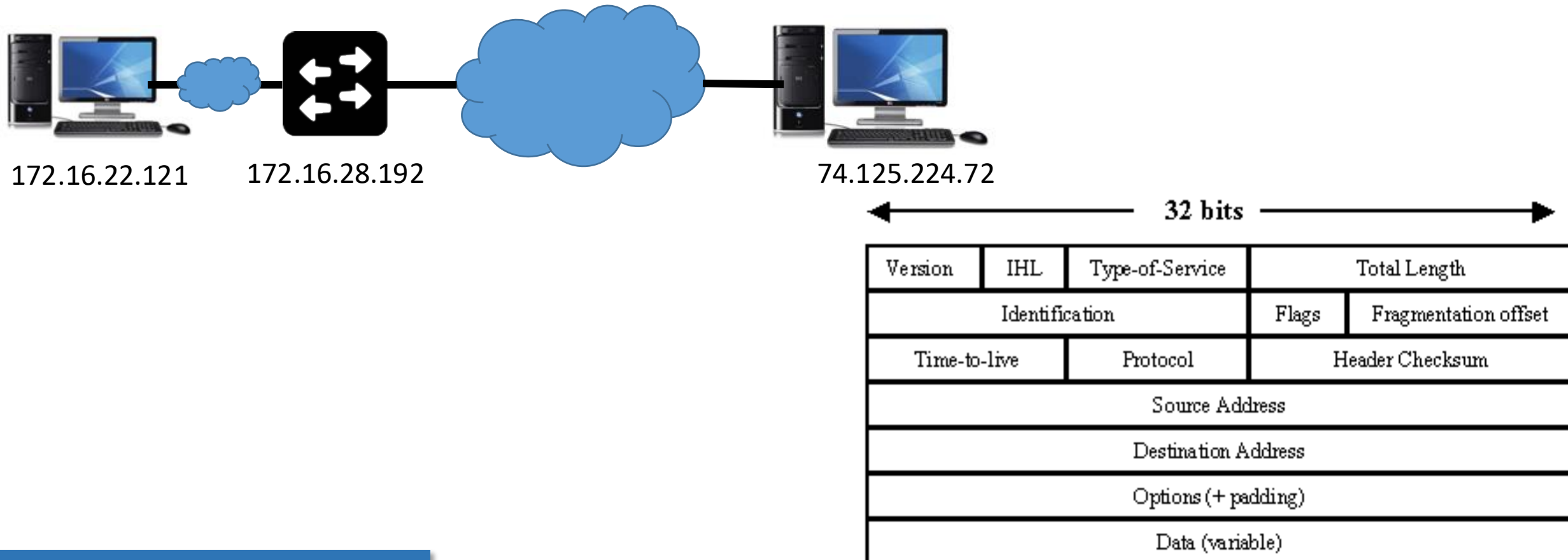
3. Network Layer: Construct the network layer packet (IP packet)

- Source IP: 172.16.22.121 (IP of your machine)
- Destination IP: 74.125.224.72 (IP obtained from DNS query)



How do you access a page at www.google.com?

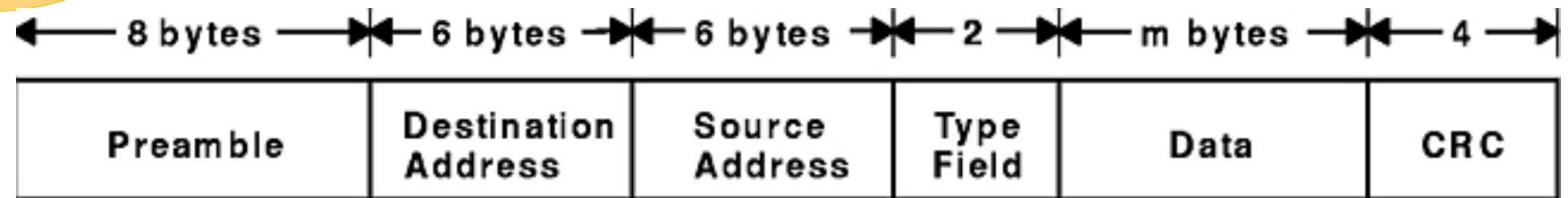
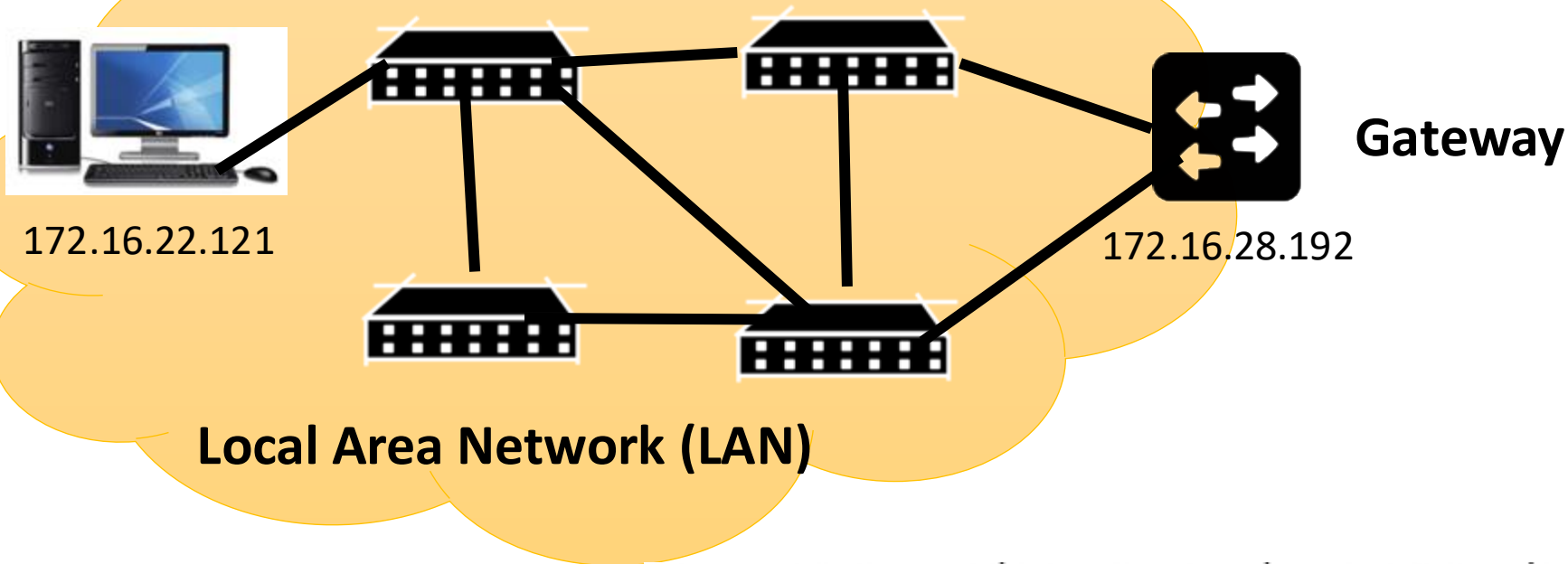
4. **Network Layer:** Use the routing procedure to find out the next hop IP to reach the Google server at 74.125.224.72. Let this IP be 172.16.28.192



How do you access a page at www.google.com?

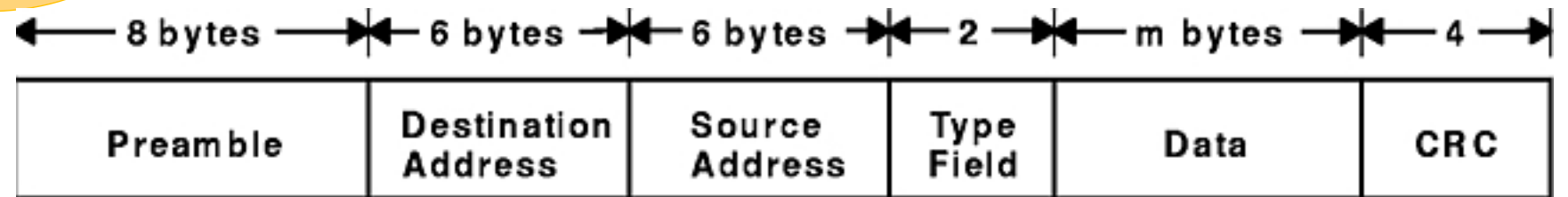
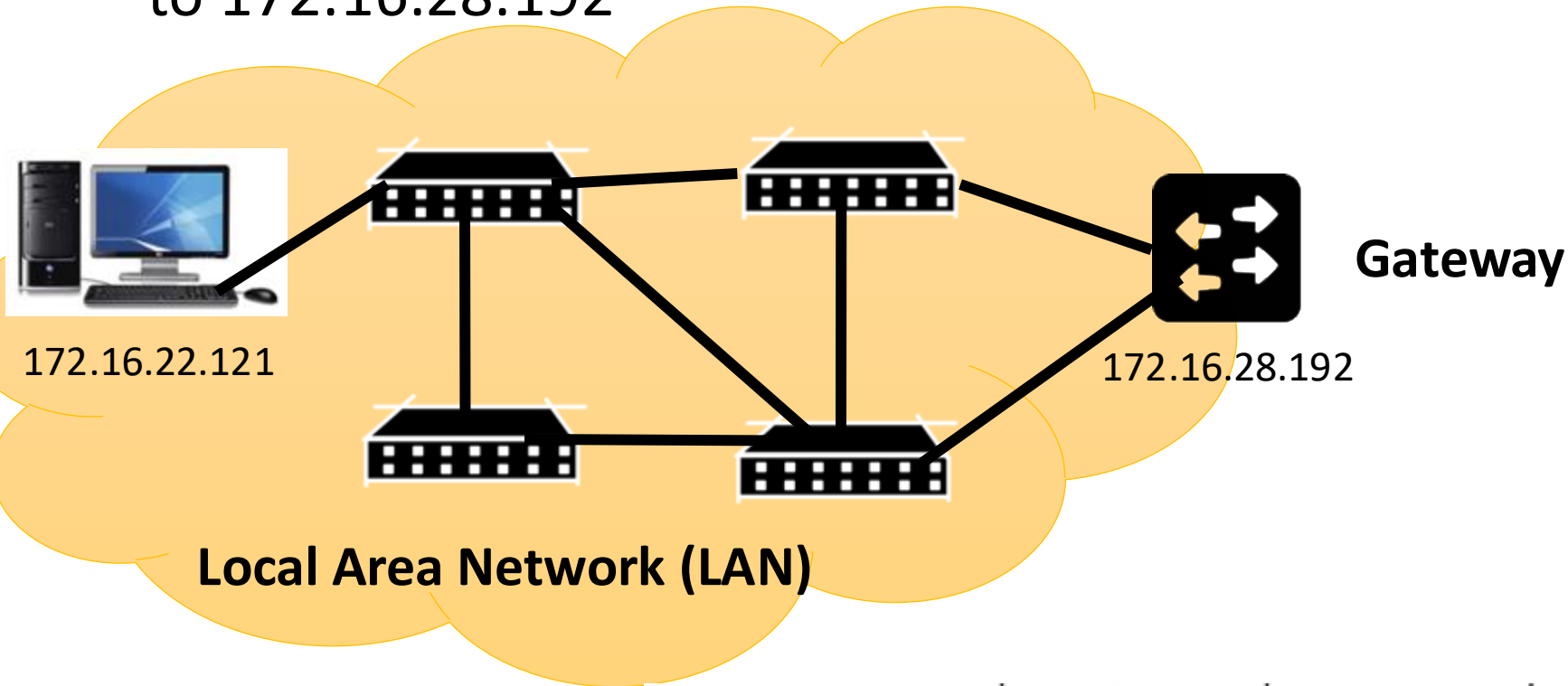
5. Data Link Layer: Construct the data link layer frame from the IP datagram.

- How do we get destination MAC address? We need the MAC address corresponding to 172.16.28.192



How do you access a page at www.google.com?

6. **Data Link Layer:** Use ARP protocol to get the MAC address corresponding to 172.16.28.192



In Summary

